Introduction

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⊙ Introduction concepts:-

* Computer Network:-

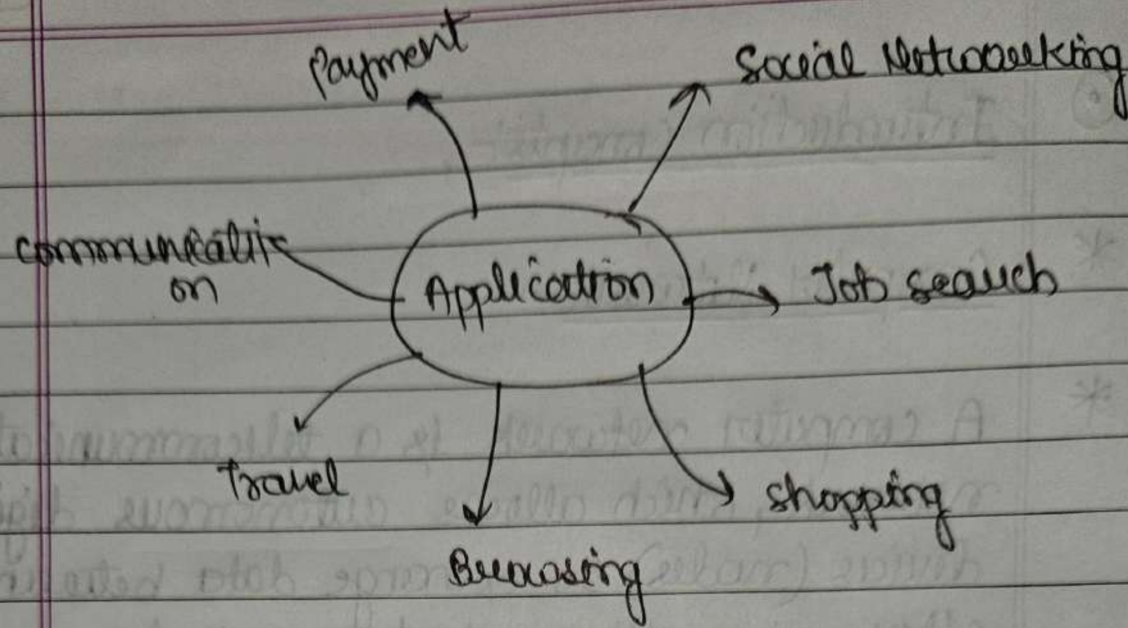
* A computer network is a telecommunication network, which allows autonomous digital devices (nodes) to exchange data between each other using either wired or wireless connections to share resources (h/w or s/w) interconnected by a single technology eg. Internet.

⊙ Goals of Network:-

1. Facilitating communication.
2. Resource sharing.
3. Data store and Access
4. Cost Efficiency
5. Reliability and Redundancy.

⊙ Application of Network:-

1. Business and commerce.
2. Education.
3. Healthcare.
4. Government services
5. Entertainment.
6. Travel and Hospitality.

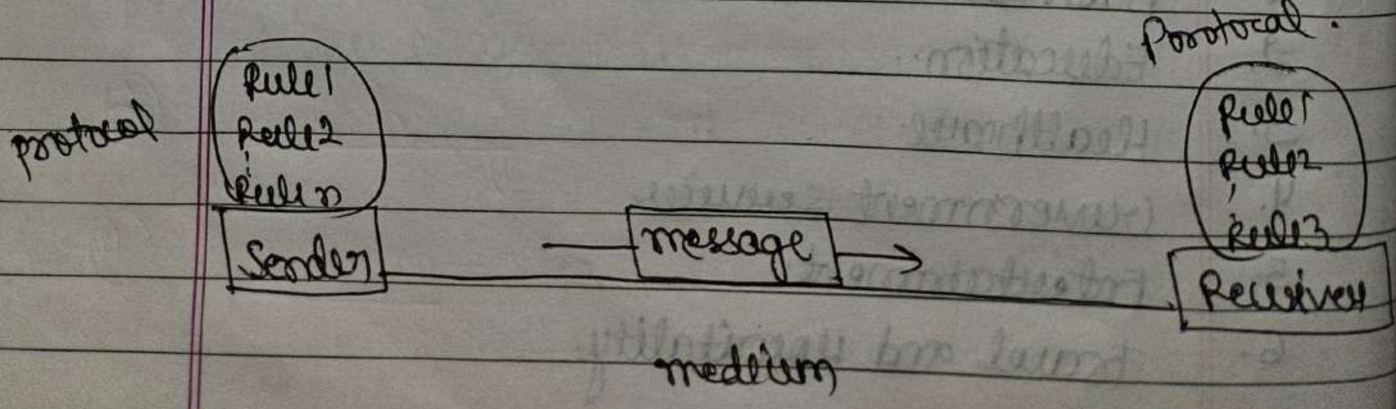


⊙ <u>Data communication:-</u>

⊙ Data communication are the exchange of data between to two devices via some transmission medium

⇉ Data communication system has five components

1. Message.
2. Sender.
3. Receiver.
4. Transmission medium.
5. Protocol.



⊙ protocol:- which includes syntax, semantics, timing
De facto, De jure.

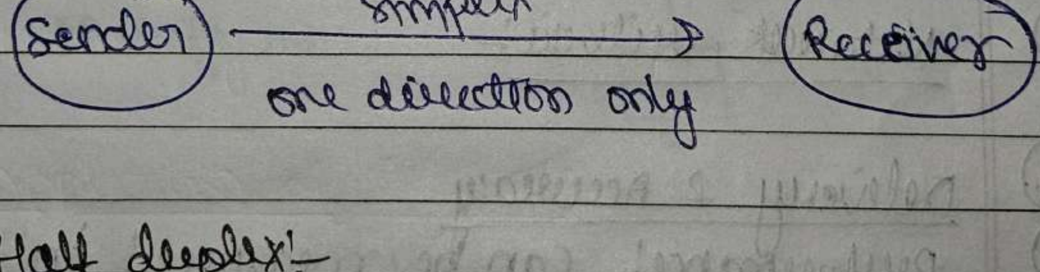
⊙ Transmission Mode:-

⊙ Data flow between two system can be categorised into three types:-

1. Simple mode
2. Half Duplex mode.
3. Full Duplex mode.

⊙ Simple mode:-

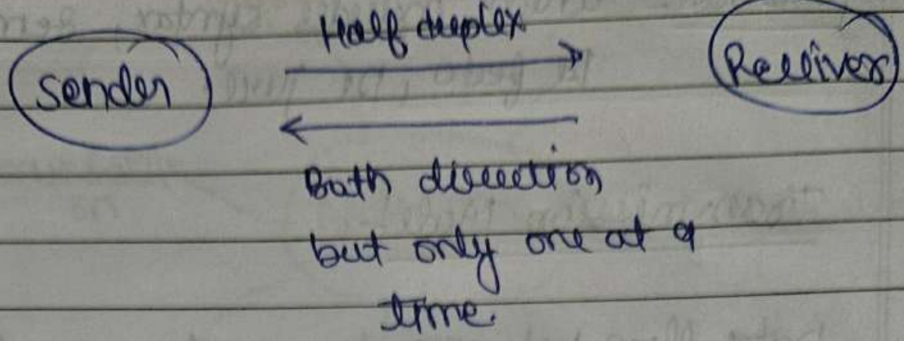
⊙ The communication is unidirectional as a one-way street
Ex. radio.



⊙ Half duplex:-

⊙ Each station can both transmit and receive, but not at the same time
Ex. 1. one lane road.
2. walkie-talkie etc.

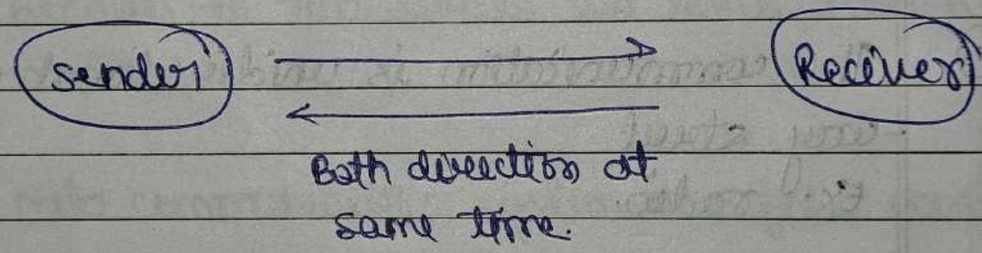
⊙ when one device is sending, the other can only receive, and vice versa.



③ Full duplex:-

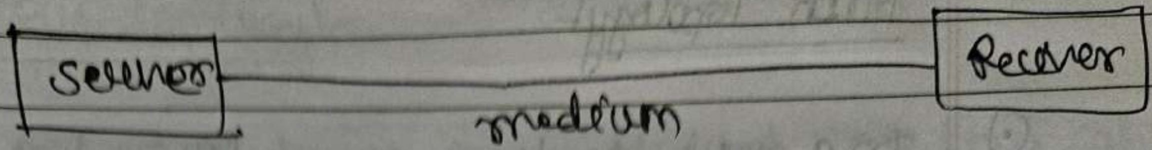
⊙ Both stations can transmit and receive at the same time. Actually it is a two half duplex connections.

Ex. telephone network.



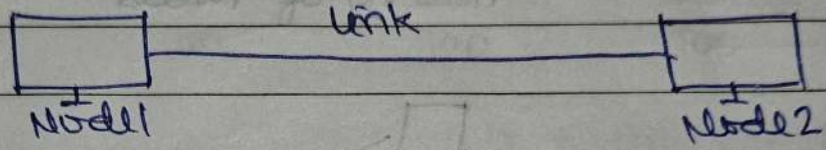
⊙ Network criteria:-

① Delivery & Accuracy
② Performance:- Can be measured in many ways including transit time, response time number of users, types of transmission medium etc.
③ Reliability :- Time taken to resolve from the failure.
④ Security:- Include protecting data from unauthorised access, protect from dama



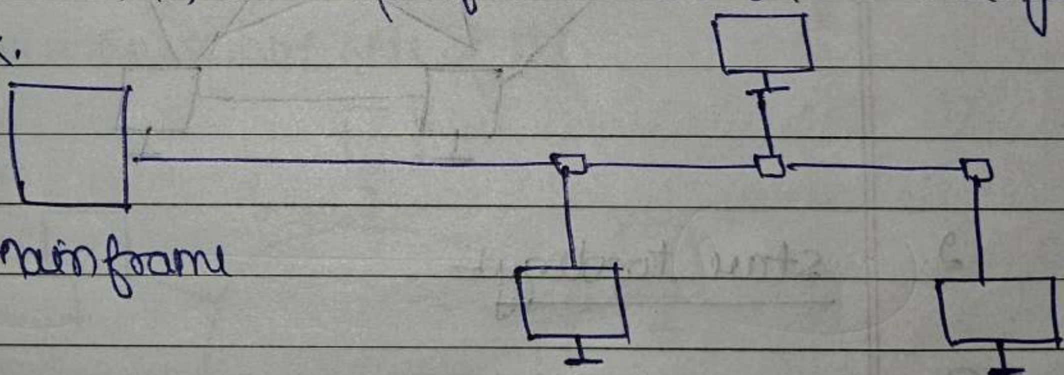
- Types of connection:-

1. point to point



2. Multipoint:-

→ more than two specific devices share a single link.



- Physical Topology:-

- Refers to the way in which a network is laid out physically. Topology of a network is the geometric representation of the relationship of all the links and linking devices to one other.

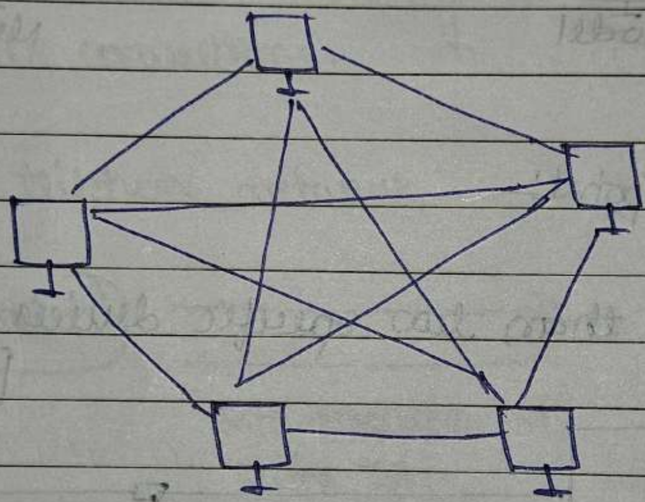
1. <u>Mesh Topology:-</u>

⊙ In a mesh topology, every device has a directed point-to-point link to every other device

cables / links = $\frac{n(n-1)}{2}$

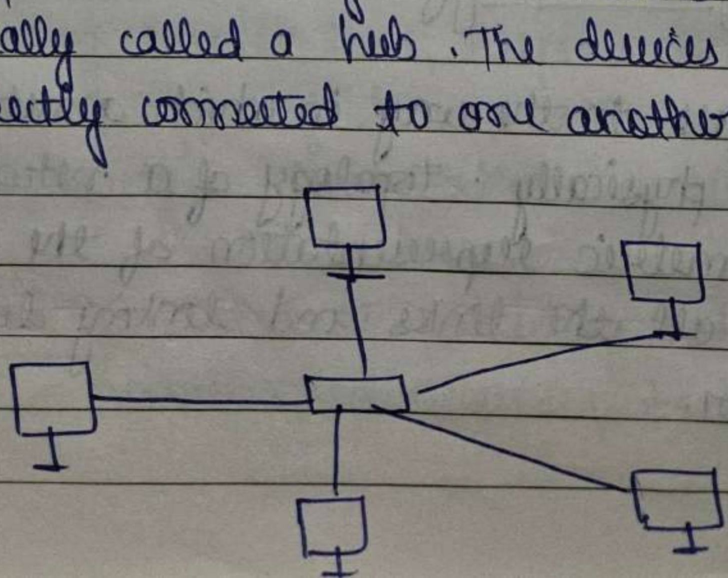
where

n = no. of nodes.



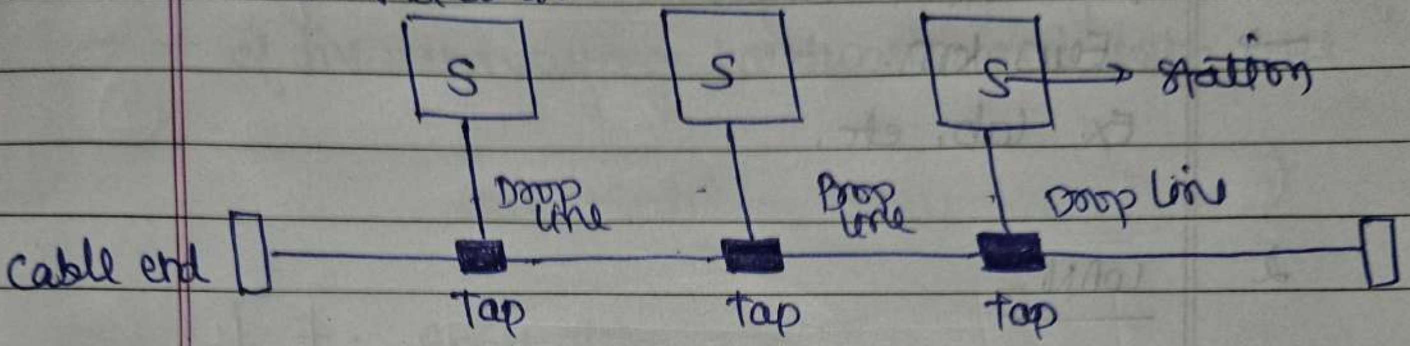
2. <u>Star topology:-</u>

⊙ In a star topology, each device has a point-to-point link only to a central controller usually called a hub. The devices are not directly connected to one another.



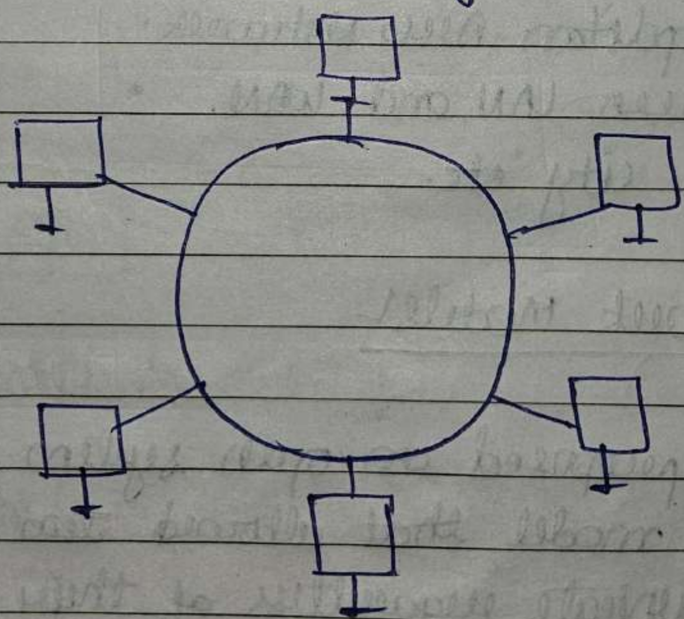
3. <u>Bus topology:-</u>

⊙ A bus topology, is multipoint. one long cable acts as a backbone to link all the devices in a network.



4. <u>Ring topology:-</u>

⊙ In a ring topology, each devices has a dedicated point-to-point connections with only the two devices on either side of it.



∴ Ring.

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LAN, MAN, WAN:

1. LAN:

- Local Area Network.
→ Few km.
Ex. lab, etc.

2. WAN:

- Wide area Network.
→ Large Area.
Ex. entire state etc.

3. MAN:

- Metropolitan Area Network.
→ between LAN and WAN.
Ex. city etc.

Network Models:

→ ISO proposed an open system interconnection (OSI) model that allowed two system to communicate regardless of their archetecture.

OSI → open source Interconnection

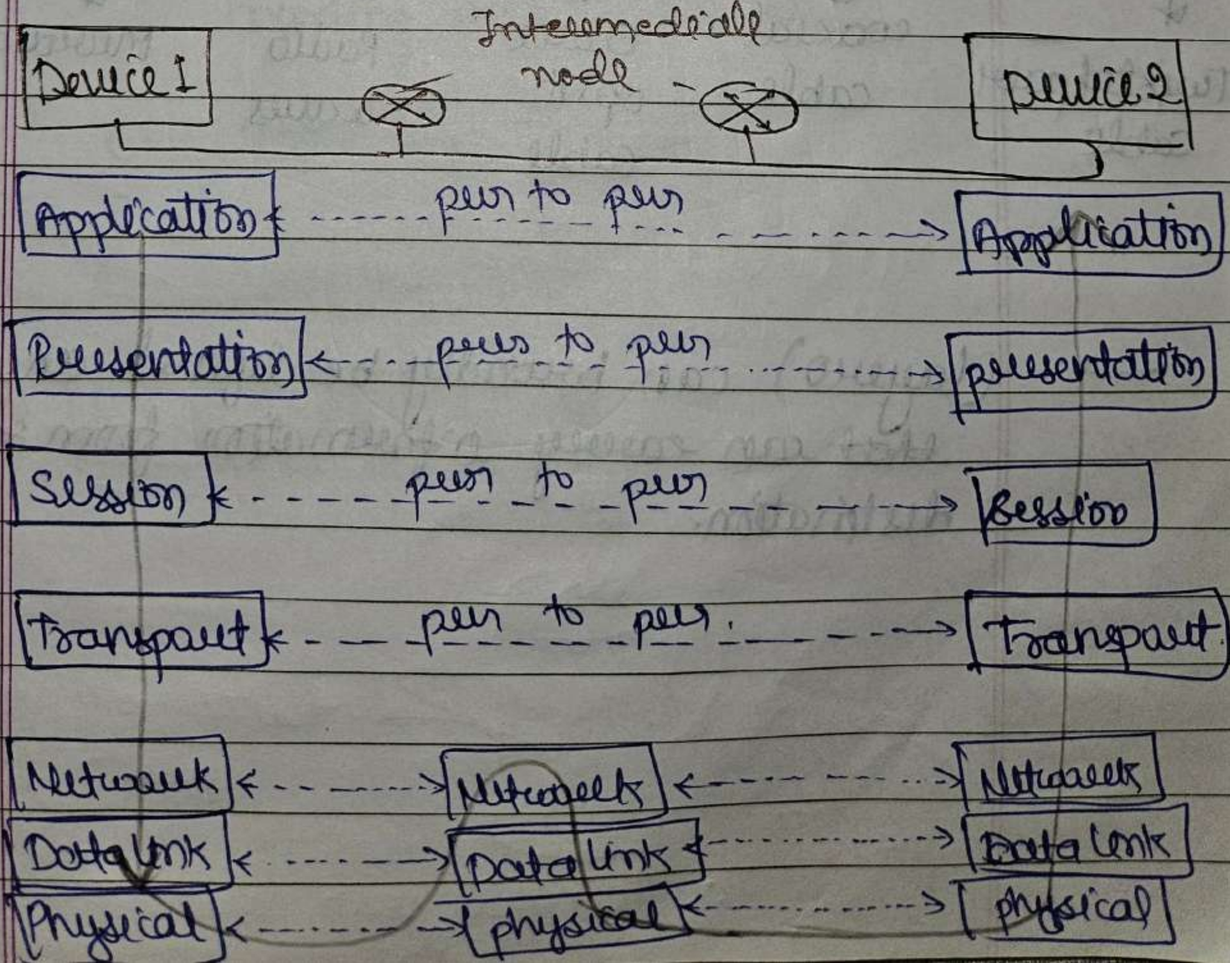
OSI:-

→ open system Interconnection
→ It facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software

OSI Model.

7. Application
6. Presentation
5. Session
4. Transport
3. Network.
2. Data Link
1. physical

layers.

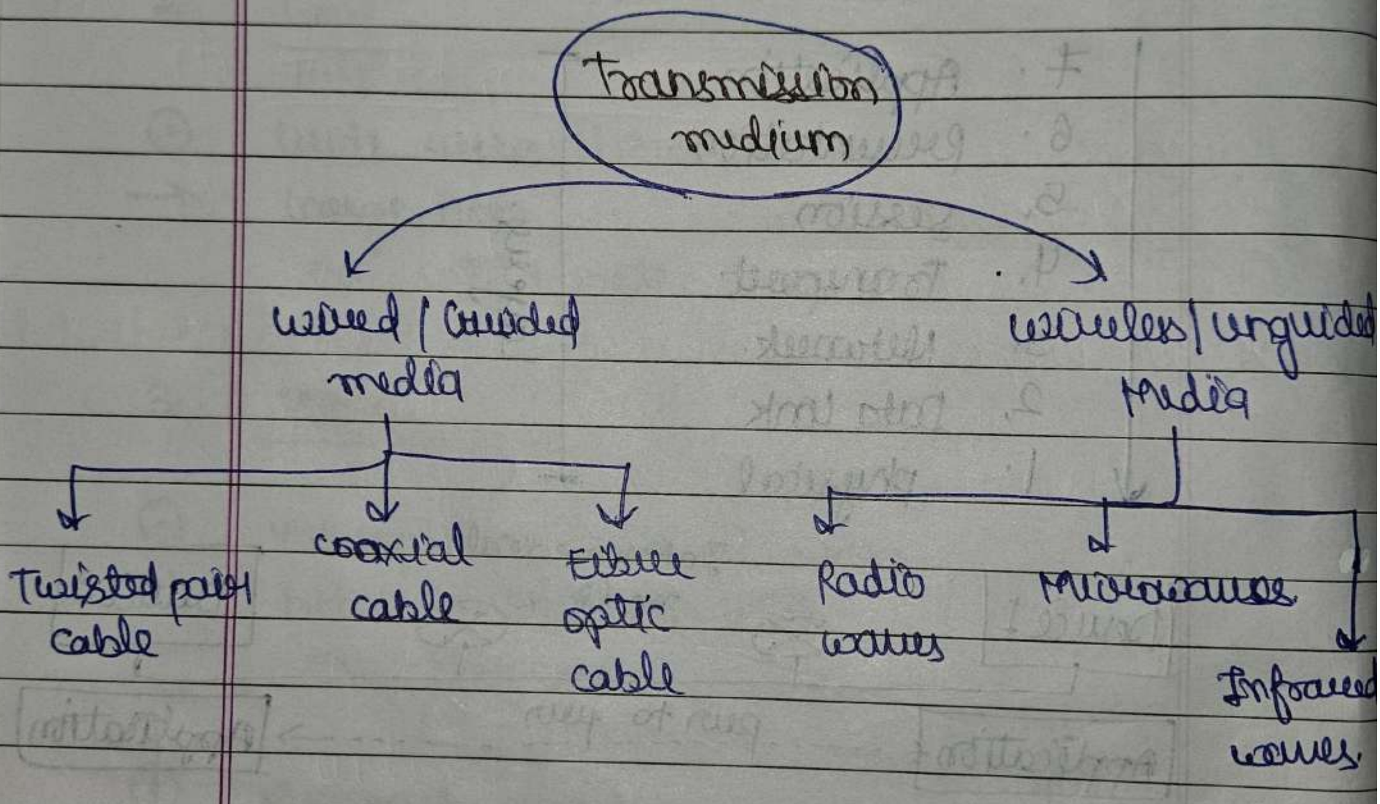


Physical Layer

1 **physical layer**

- Thy physical layer data consists of a stream of bits (sequence of 0s or 1s) with no interpretation To be transmitted, bits must be encoded into signals - electrical or optic

- Transmission media:-



→ (layer 0) can broadly be defined anything that can carry information from source to destination.

Thearem

⊙ Shannon- Haritley Theorm :-

$$C = B \log_2 \left[1 + \frac{S}{N} \right]$$

where,

B= Bandwidth of channel.
S= Signal power.
N= Noise power.

Que. A channel has Bandwidth 4K z and a signal to noise ratio is 30 dee. Determine the maximum information ratio in 128 lable of Encoding.

Given:-

Bandwidth B= 4KHz = 4000Hz
Signal to noise ratio = 30
Encoding with = 128 levels.

$$C = 4000 \log_2 [1+30]$$

$$C = 4000 \log_2 (31)$$

$$\boxed{C = 1.98 \times 10^4 \text{ bps.}}$$

① **Wired media:**

⊙ A data communication & computer network, wired media refers to physical paths (cables/wires) through which data signals are transmitted.

⊙ **Types:**

(i) **Twisted pair cable:**

⊙ Consist of two conductors (copper), each with its own plastic insulation, twisted

→ Consists of pairs of insulated copper wires twist together.

→ Twisting reduces interference (crosstalk).

Twisted pair cable

- unshielded twisted pair
- shielded twisted pair

→ Twisted pair cable

∴ twisted pair cable.

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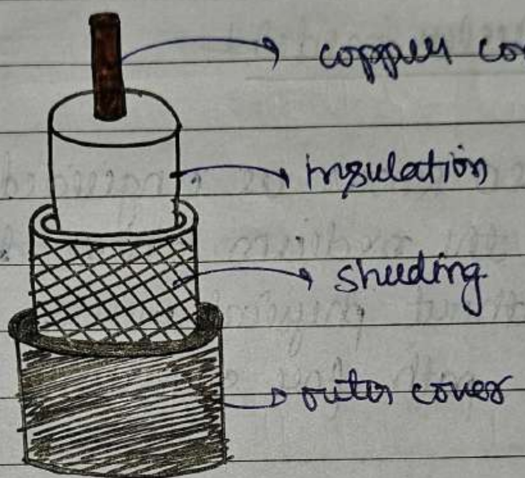
(ii) Coaxial cable

- Has a central copper conductor, surrounded by insulation, shielding and an outer cover.

→ provide better noise resistance than twisted pair.

→ used in cable, TV, broadband internet.



∴ coaxial cable.

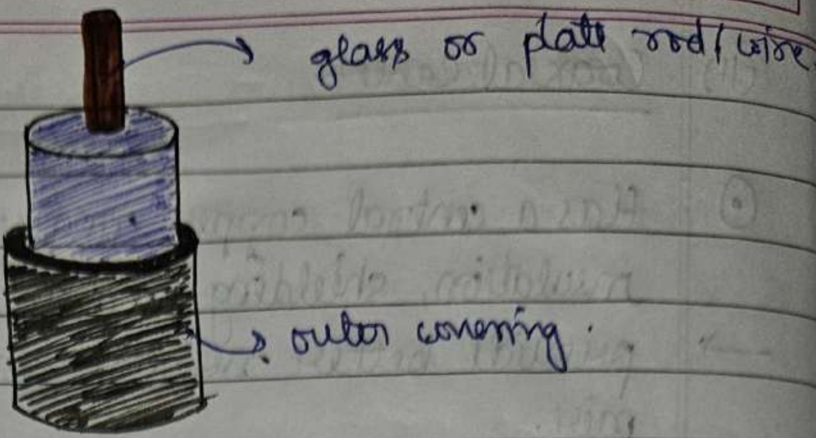
(iii) optical Fiber cable

- uses light signals (lasers or LEDs) instead of electrical signals

→ core is made of glass or plate.

→ provides → (i) high bandwidth.
(ii) long-distance transmission.
(iii) immunity to electromagnetic interference.

→ used in backbone networks, ISPs and high speed internet.



∴ optical fiber cable.

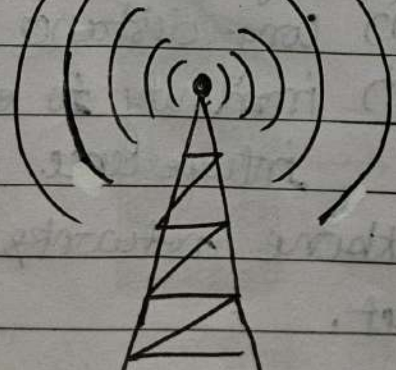
② **wireless media :-**

→ Also known as unguided transmission media in the medium where data is transmitted without physical wires, using air or space a the path for signals.

⊙ Types :-

(i) Radio waves :-

→ Frequency range 3KHZ - 1GHZ.
→ can pass through walls
→ used in radios, Bluetooth, wi-fi etc.



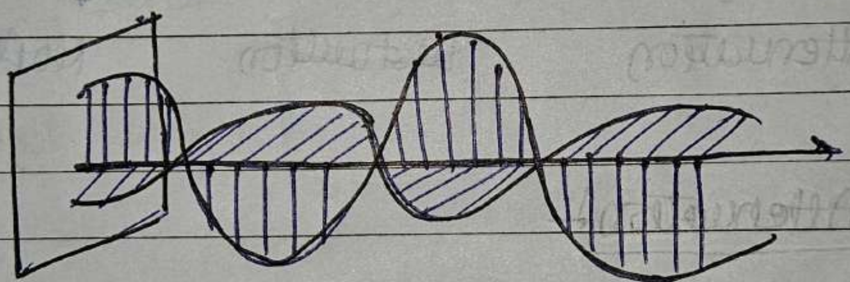
∴ Radio waves

(ii) Microwaves:-

→ Frequency range:- 1 GHz - 300 GHz.
→ Need line of sight.
→ used in satellite communication, cellular networks, WiMAX.
→ High data transmission rate.

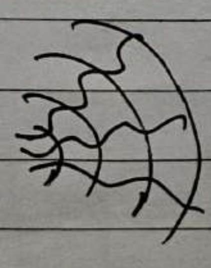
Types
- Terrestrial MW
- Satellite MW.

Ex:- mobile phones, TV transmission, radar, etc.



(iii) Infrared:-

→ Frequency range:- 300 GHz - 430 THz.
→ cannot pass through walls.
→ works only in short range and line-of-sight
→ Remote control, wireless mouse / keyboards etc.



∴ Infrared.

Signal impairment (transmission)

- Signals travel through transmission media which are not perfect.
  The imperfection causes signal impairment.

Note: means signal at beginning is not same at the end.

Causes of Impairment

Causes

- Attenuation
- Distortion
- Noise.

Signal Transmission

Digital signal — Anology signal

⊙ Signals:-

⊙ Signal is an information that is converted into electrical form which is suitable for transmission.
It is used to carry data from one device to another.

* Types:-

1. Digital signal.
2. Anology signal.

⊙ Digital signal:-

⊙ It is non-continuous, discrete signal that carries data in binary form i.e 0 and 1.

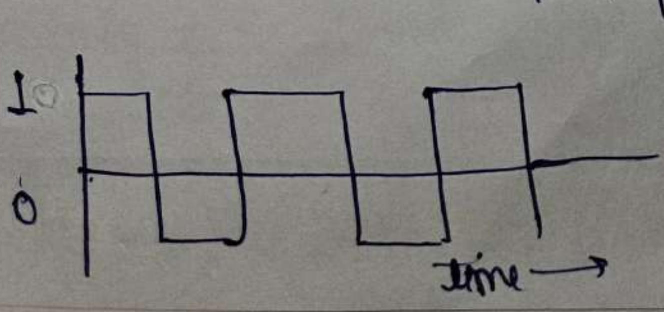
⊙ It is defined using bit rate and bit interval.

1. Bit interval:- Time required to send signal bit.

2. Bit rate:- Frequency of bit interval.

⊙ Transmits data in 0s and 1s.

⊙ Immune to noise and doesn't face distortion.



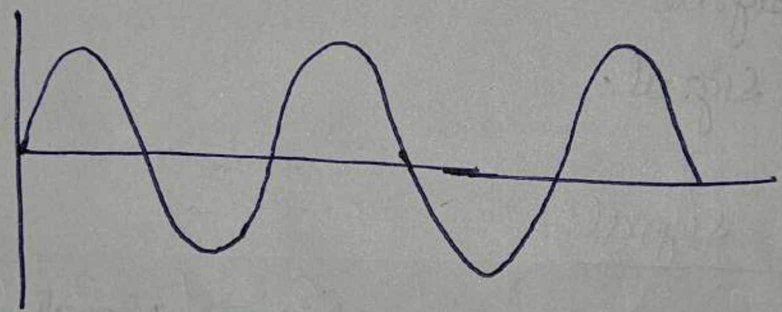
Analog Signall

- A continuous wave form that change over the time.

Typest
- Simple A.S
- Composite A.S

- Analog signal is defined using factors like.
  1. Amplitude.
  2. Frequency.
  3. Phase

- The signal is not immune to noise, thus can face distortion



- There is no fixed range.

⊙ Line coding schemes

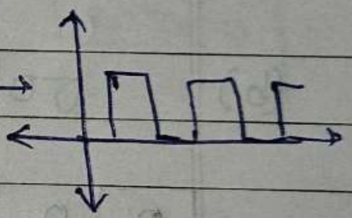
⊙ It is the process of converting binary data, sequence of bits to a digital signal (voltages or light pulses) that can be transmitted over a medium.

→ **Classification of line coding scheme:-**

Unipolar — Polar — Bipolar.

Ex:-

| 0 | 1 | 0 | 0 | 1 | 1 | 1 | → Line coding →



⊙ Properties of line codes:-

1. Bandwidth used is reduce
2. Power is efficiently used.
3. Probability of error is reduced.
4. Error detection and correction capabilities.

⊙ Unipolar:-

→ Uses only one polarity of voltage.

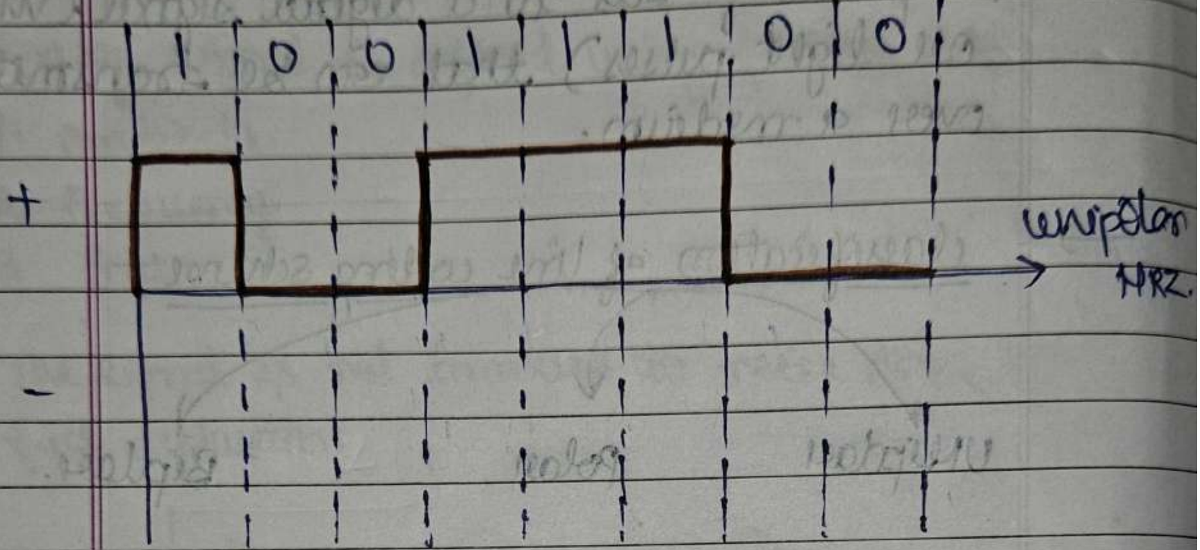
→ 0V → 0
+V → 1

→ Simple but not efficient.

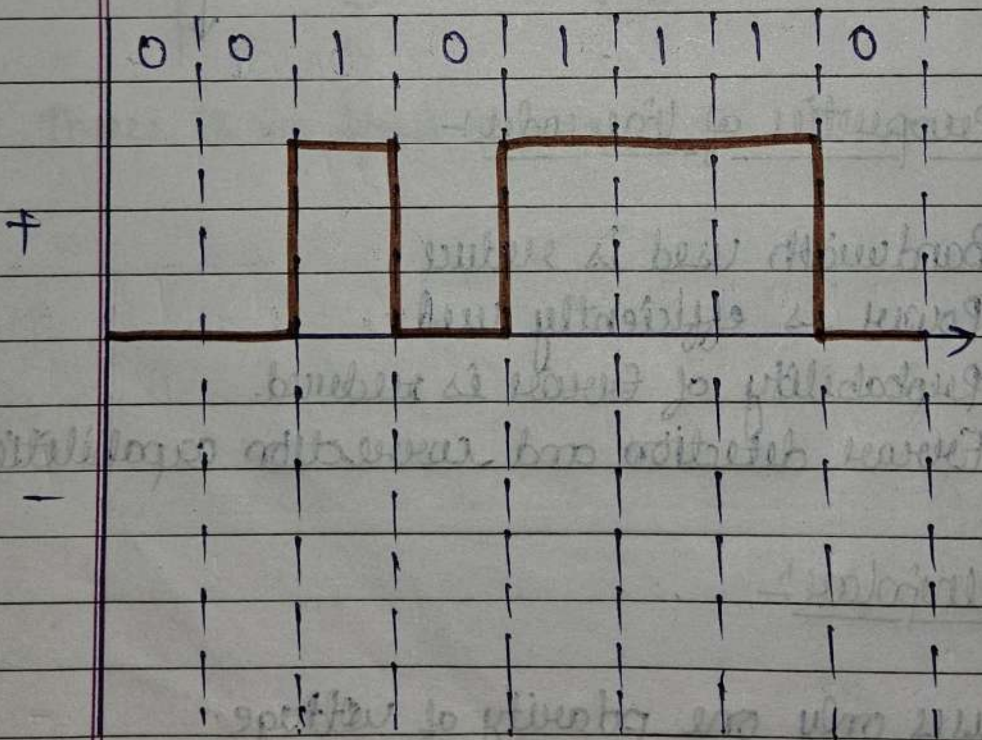
(i) Digital Data -

∴ [1- up
0-down]

(a) 10011100.



(b) 00101110



⊙ Polar :-

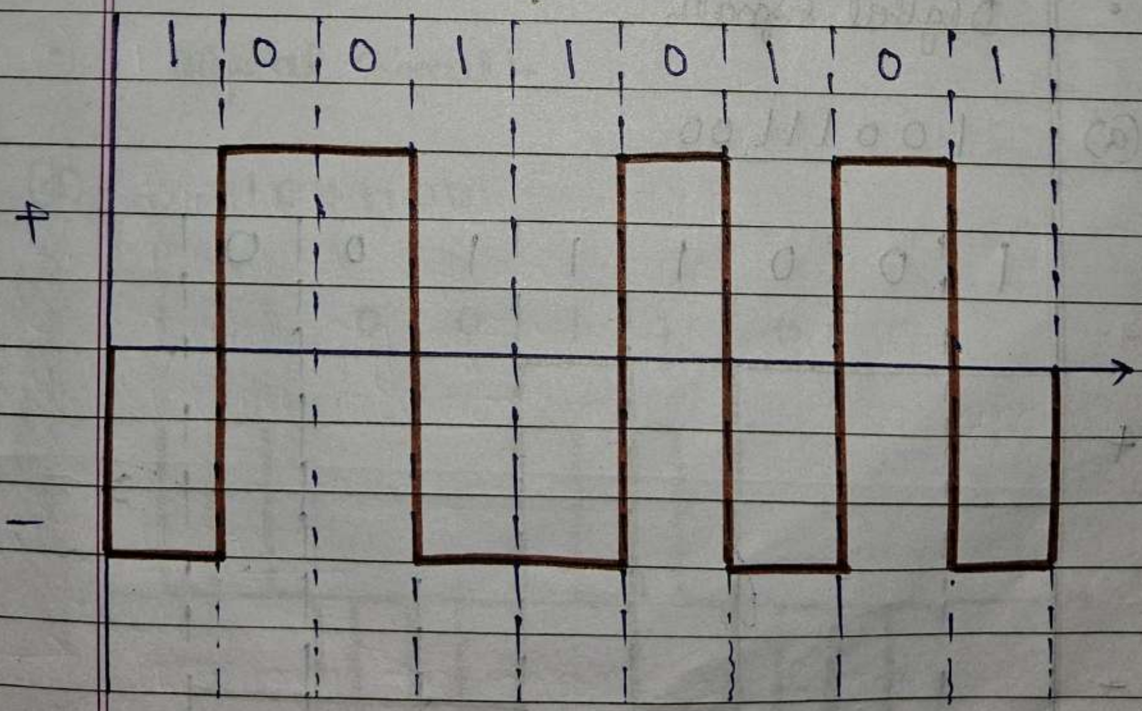
→ A technique to represent digital data (1s and 0s) into signals for transmission.

Polar
- polar NRZ-L
- polar NRZ-I.
- RZ (return to zero)

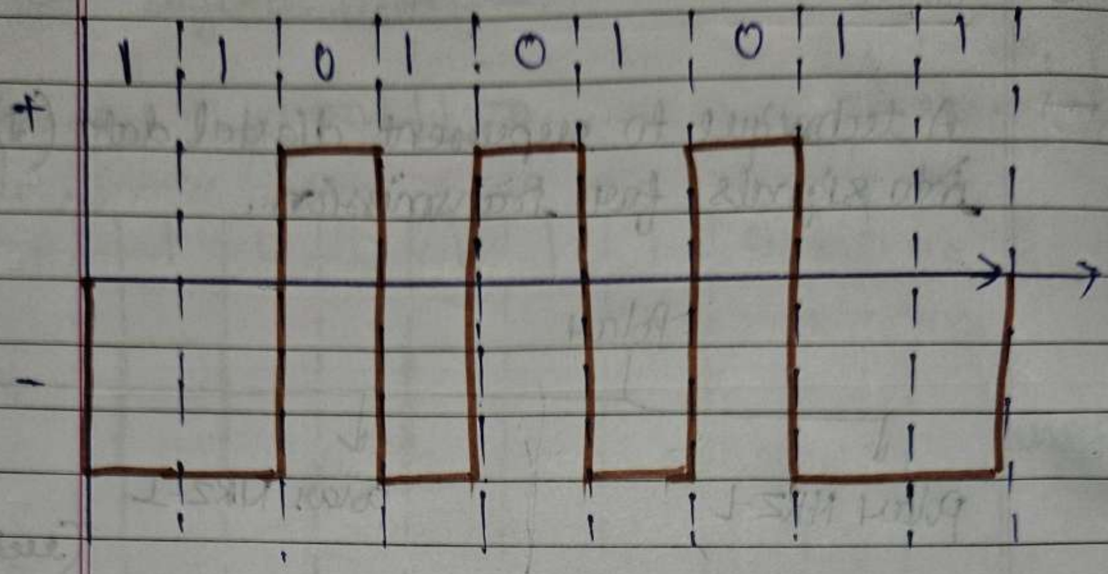
(1) Polar NRZ-L :-

• Digital data :- [1- Zero. 0- Hero.]

(a) 100110101



(b) 11010011

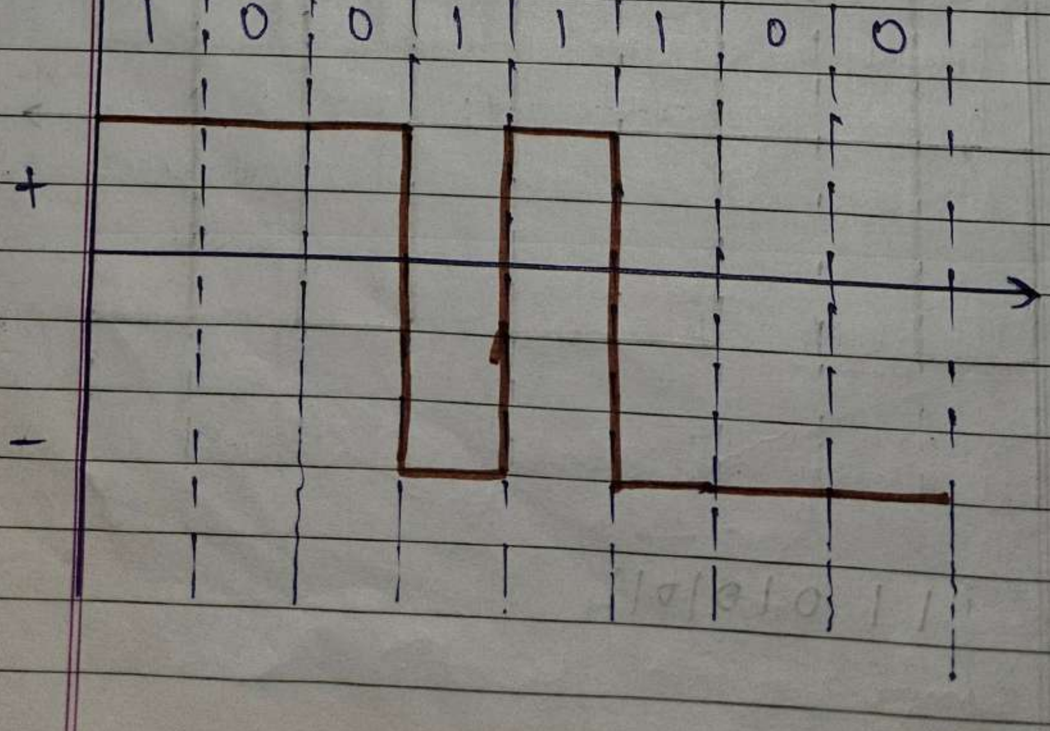


② Polar NRZ-I

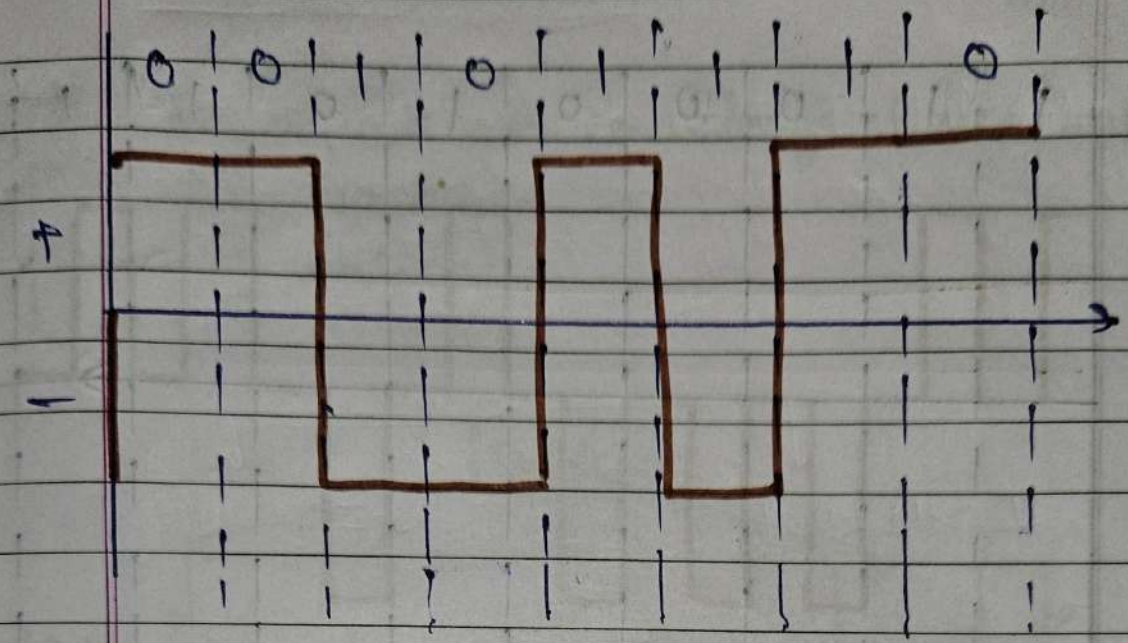
→ 0 – No transition
→ 1 – Transition.

• Digital signal

(a) 10011100

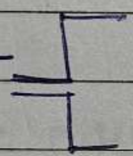


(b) 00101110



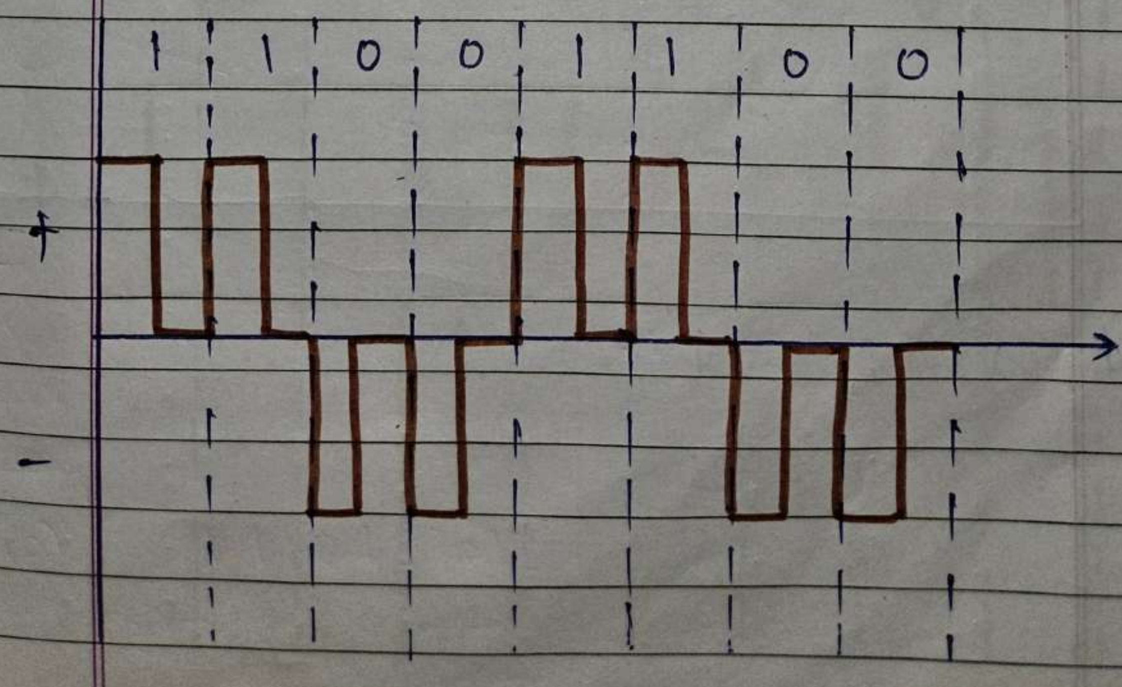
(3.) RZ - (Return to Zero)

→ 0 -
→ 1 -

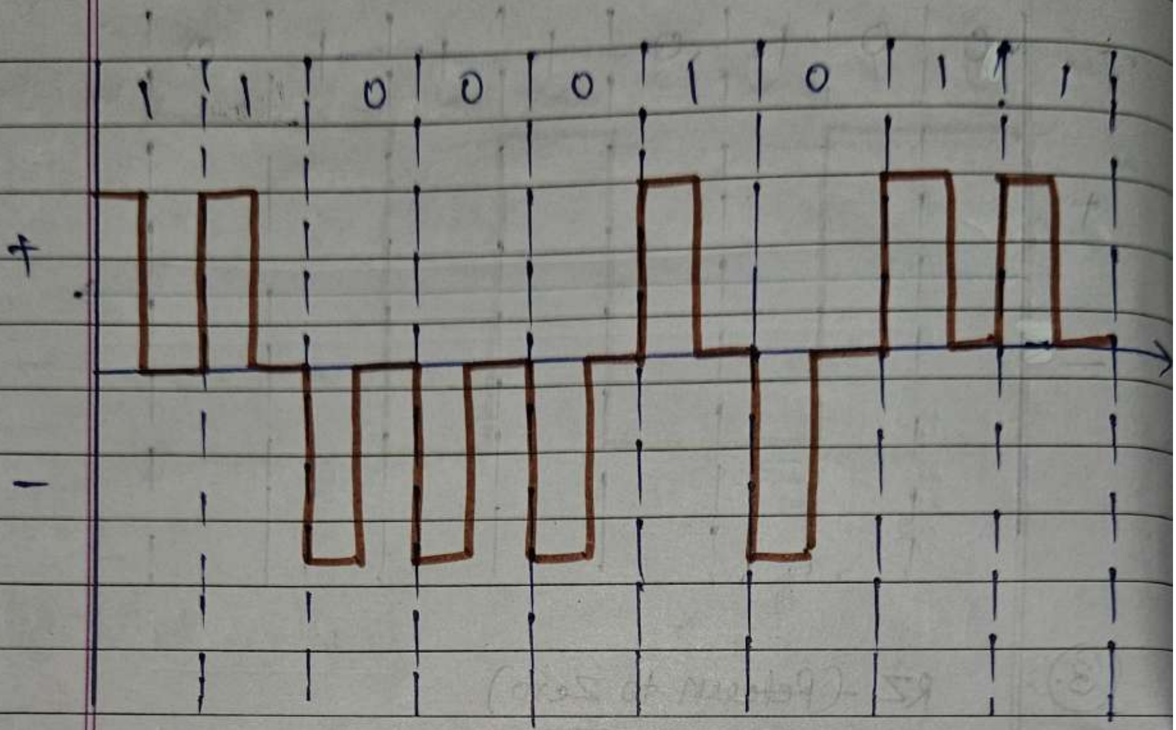


• Digital signal :-

(a) 11001100



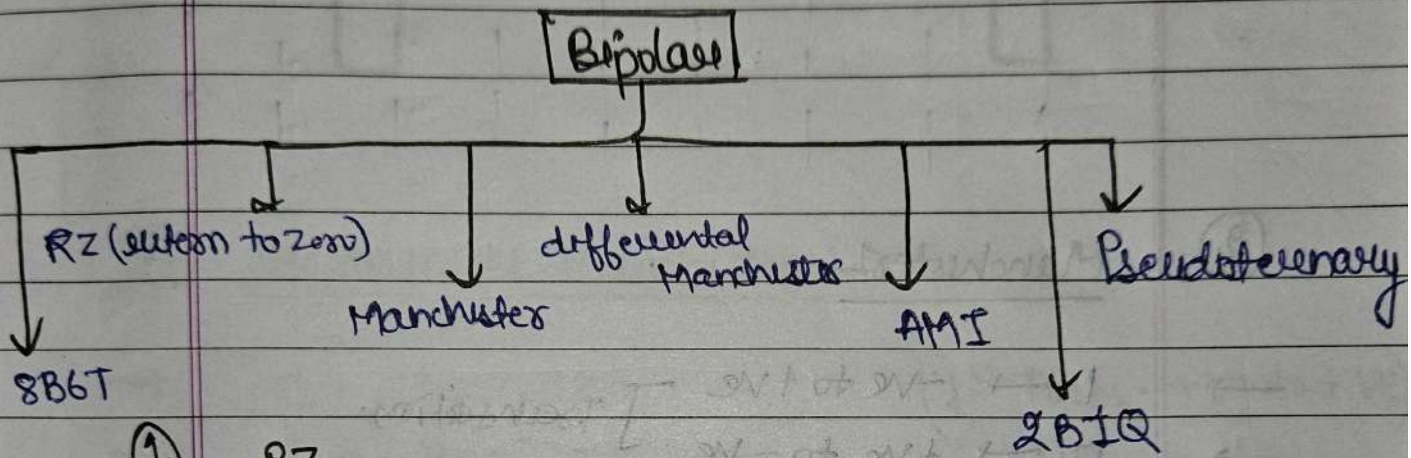
(b) 110001011



⊙ **Bipolar :-**

→ In Bipolar three voltage levels are used.

Voltage
- +V (positive)
- 0 (zero)
- −V (negative)

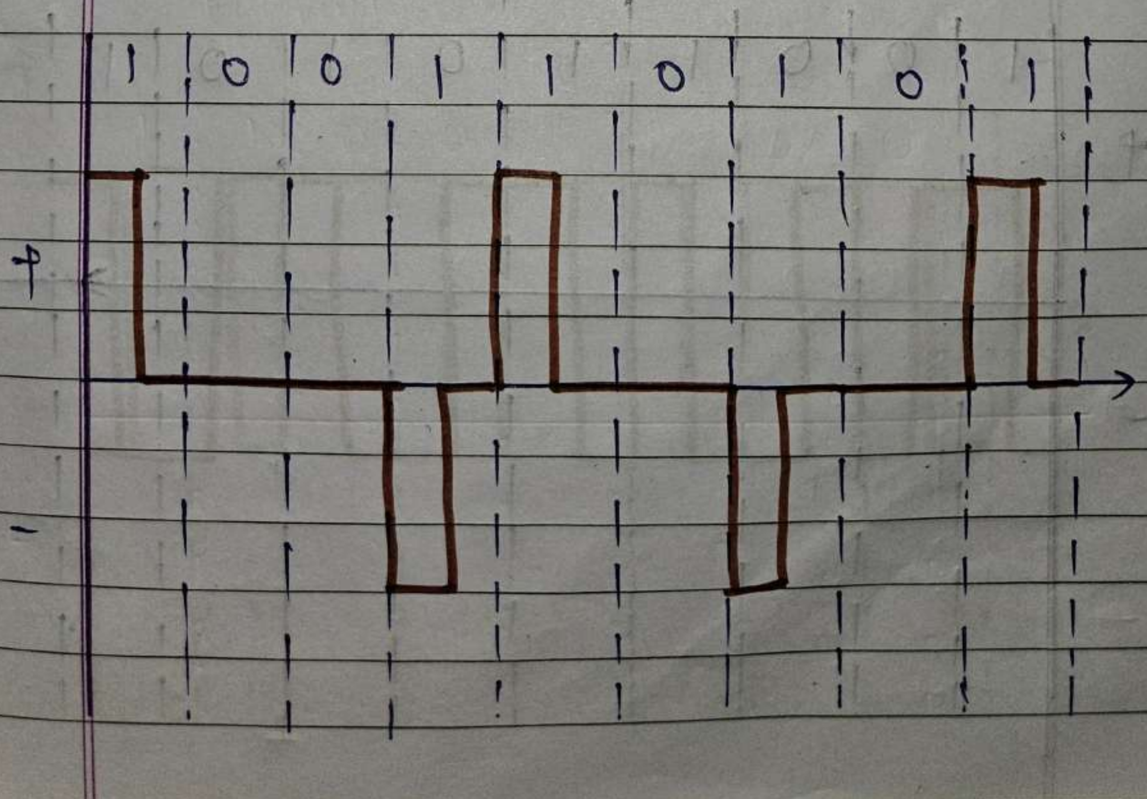


① RZ

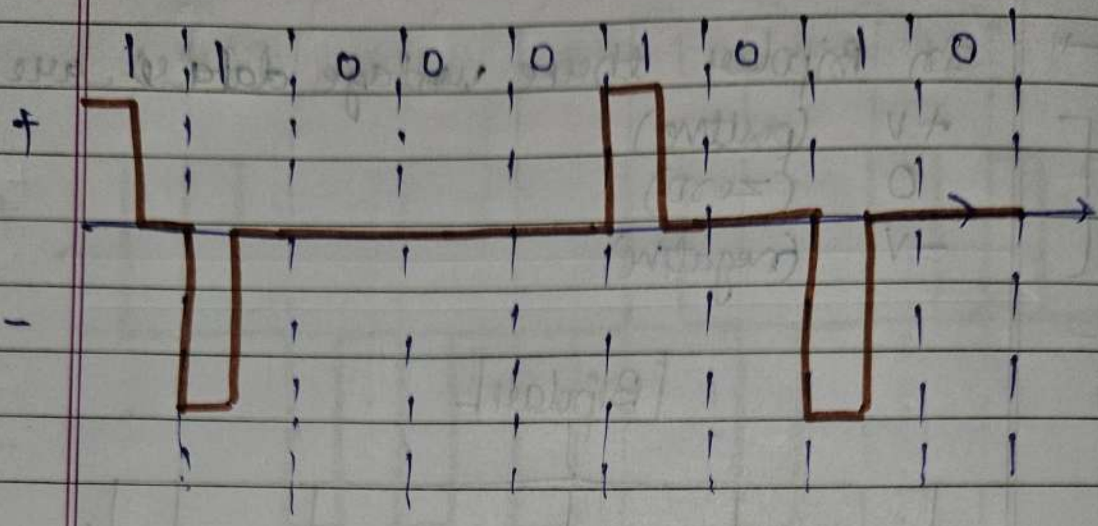
→ change direction of I.

• Digital Data

(a) 100110101



(b) 11 0001010

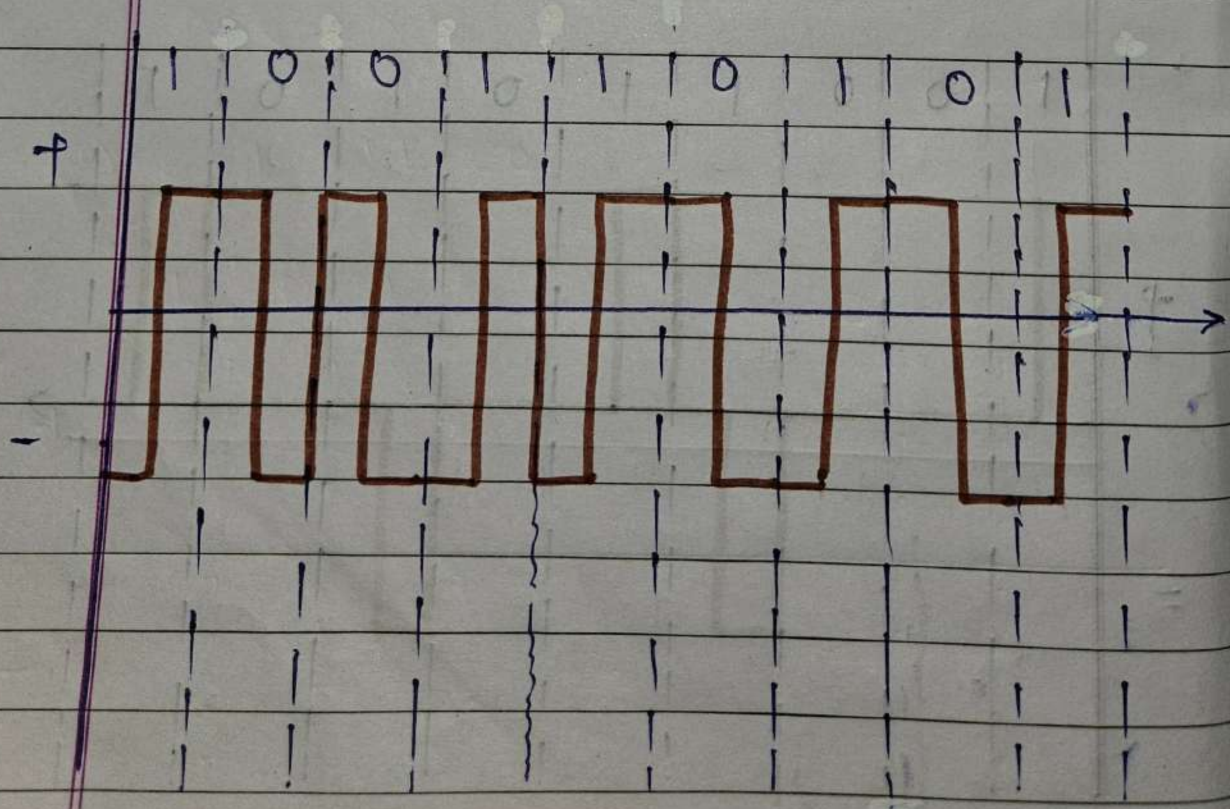


② Manchester

→ 1 → -ve to +ve] Transition.
→ 0 → +ve to -ve

• Digital Data :-

(a) 10011 0101



(b) 10011100

+

-

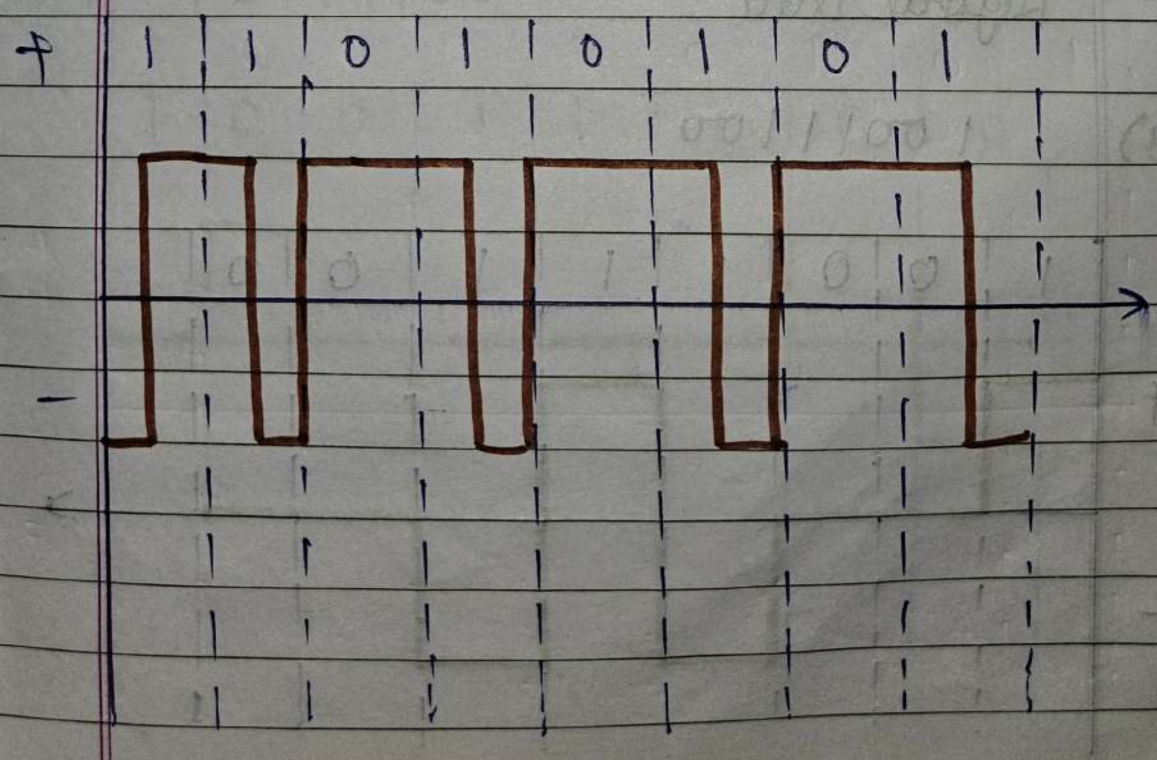
③ Differential Manchester:-

→ 0 - Transition.
→ 1 - No-transition.

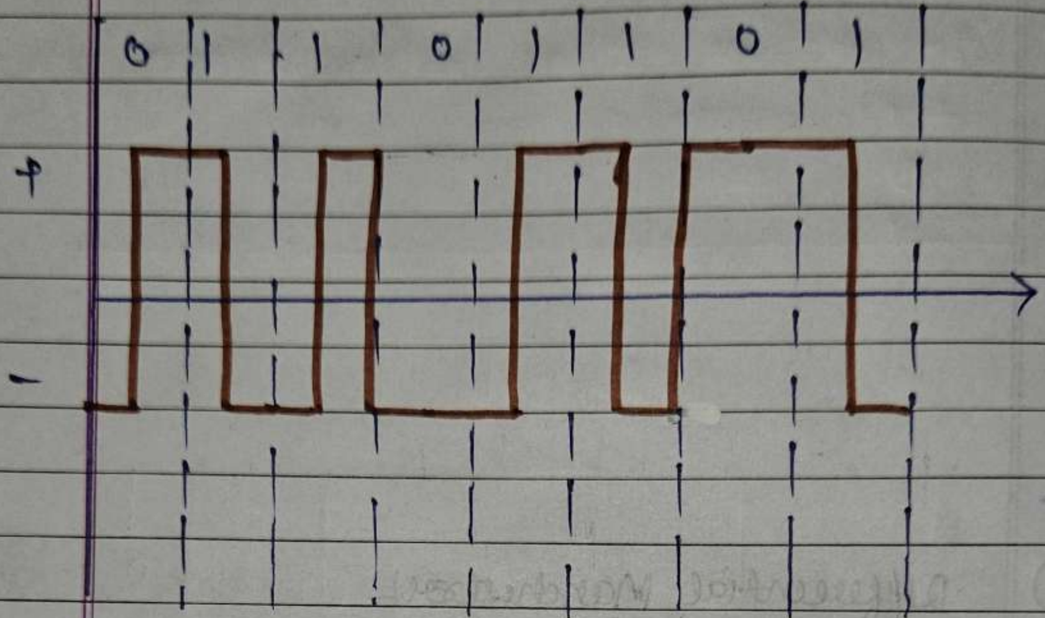
(1st Trans. -ve to +ve)

• Digital code:-

(a) 11010101



(b) 01101101

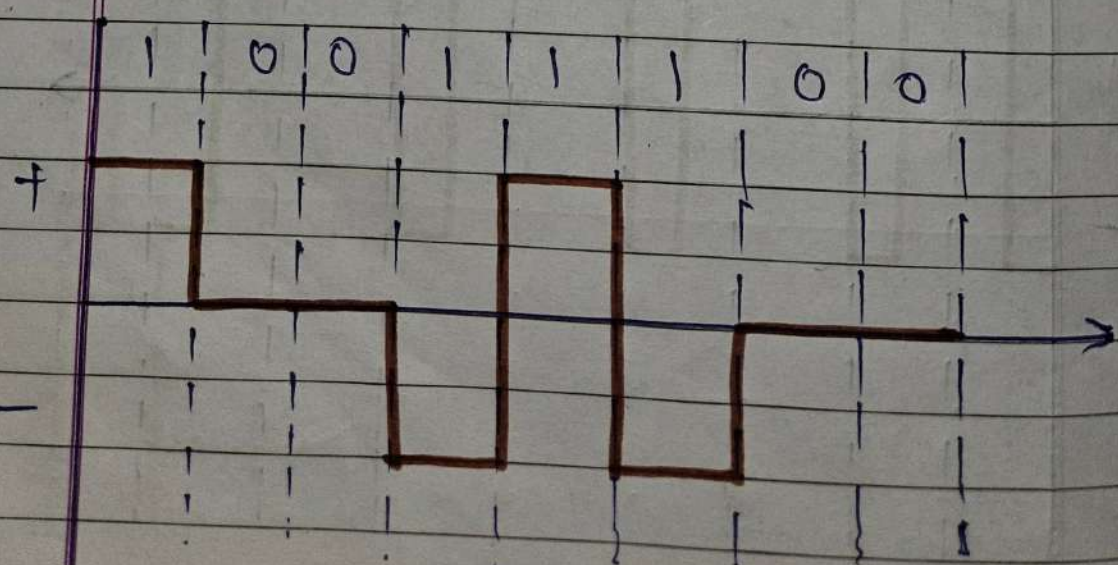


④ AMI:-

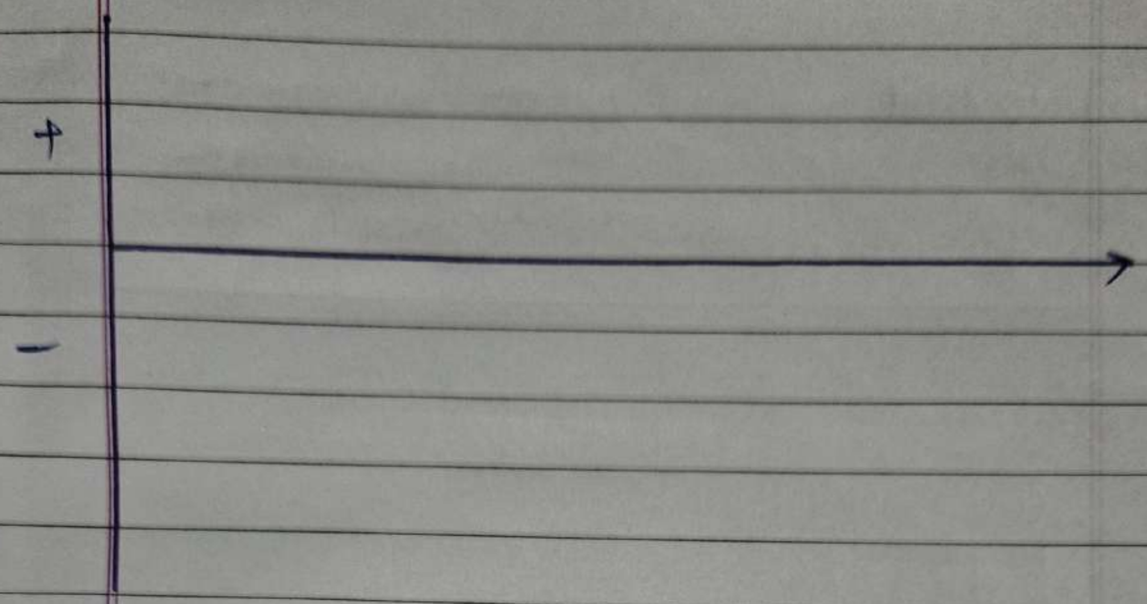
→ 0 → Draw on reference line.
→ 1 → Draw 1 line above and next line below

• Digital Data

(a) 10011100



(b) 0010110

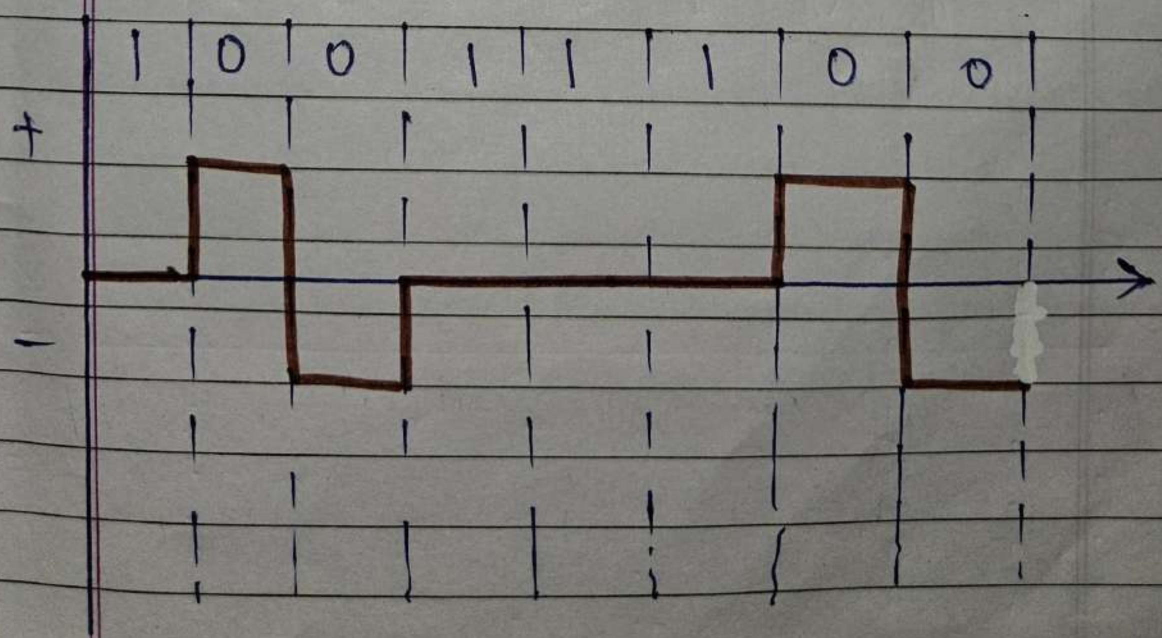


⑤ <u>Pseudoternary :-</u>

→ 1 – Draw on reference line.
→ 0 – Draw 1 line above and next line below.

• Digital Data.

(a) 10011100



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Switching Methods:-

1. Circuit Switching.
2. Packet switching.

- Switching is a process to forward data packets coming in form one port to another port.

Circuit switching:-

- Circuit switching is a method where a dedicated communication path or channel is established between the sender and receiver for the entire duration of communication session.

Packet switching:-

- It is a way of sending data over a network by breaking it into small pieces, called packets.

Data Link layer

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⊙ **Data link layer:-**

⊙ The data link layer divides the stream of bits received from the network layer into managable data units called frames.

⊙ **Error Detection and correction:-**

⊙ **Error**

⊙ When data is transmitted from one device to another device,

$$\left[\begin{matrix}\text{send} \\ \text{data}\end{matrix} \neq \begin{matrix}\text{receive} \\ \text{data.}\end{matrix}\right] \text{ error.}$$

→ The external noise can changes bits from 1 to 0 / 0 to 1 in binary information.

→ This change in values changes the meaning of the actual message and is called an Error.

⊙ Detection
- → Block coding
- → Parity
- → check sum
- → CRC (cyclic Rotating check

⊙ Correction
- → Hamming code

(2) Parity:-

(•) The parity bit is extra bit added to the message from sender side, to help in error detection at the receiver side.

→ Also called as Data-word bits are codeword.

1. **Even parity:-** If numbers of 1's is even in data bits, parity bit value is 0.

2. **Odd parity:-** If numbers of 1's is odd in data bits, parity bit value is 1.

Ex:-

1	0	1	0	1	0	1	0

→ parity bit (∴ even)

0	1	1	0	1	1

→ parity bit (∴ odd)

(•) Limitations:-

1. Not suitable for detection of multiple errors.
2. Cannot reveal the location of erroneous bit.

(3) Checksum:-

- It detects all errors involving an odd numbers of bits as well even number of bits.

[Sender Check Sum] ≠ [Receiver Check sum] [∴ Error]

→ **Sender:-**

1. Divide data into X sections also every section must be of Y bits.
2. Performing checksum followed 1's complement
3. Send data after adding checksum in it.

→ **Receiver**

1. //
2. Perform binary addition followed by 1's complement.
3. If result is all zero then there is not any error or else Error is detected.

Ex:-

$\overbrace{\text{00001001}}^{\text{8 bits}}$	8	8	8	8	Check sum
00001001	A	B	C	D	-21
↓ 9	↓ 2	↓ 4	↓ 5	↓ 1	

⇒ $9+2+4+5+1 = 21$

⇒ 1's complement = -21.

Send to receiver

=) Same process done by receiver.

If zero then ok otherwise error.

(4) CRC - Cyclic Redundancy check :-

→ Based on binary division.

→ Can detect all odd errors, single bit, burst errors of length equal to polynomial degree.

Ex:-

1010101010

⇒ given poly. $x^4 + x^3 + 1$. (Divisor)

no. of zeros added.

⇒ $1 \cdot x^4 + 1 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x^1 + x^0 \cdot 1$.

⇒ 11001 (Cofficient).

XOR

0	0	→ 0
0	1	→ 1
1	0	→ 1
1	1	→ 0

```
11001 ) 10101010100000 ( 1111
        11001
        011000
         11001
         000011010
             11001
             00011000
                11001
                000010
```

Take and append it.

in [10101010100010]

→ If at the receiver side the divisor will be 0 if not, then error.

⇒ efficiency $= \left(\frac{10}{14} \times 100\right)$.

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- Flow control:-

- It refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgement.

error control → In data link layer is based on automatic repeat request, which is the retransmission of data.

Error control.

protocol
- Noiseless channels
  - Simplest
  - Stop-and-wait
- Noisy channels
  - Stop-and-wait ARQ
  - Go-Back-N-ARQ
  - Selective Repeat ARQ

- Flow control Techniques:-

1. Stop-and-wait protocol.
2. Sliding window protocol.

- Stop-and-wait protocol:-

- The idea of stop-and wait protocol is straight forward.

→ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.

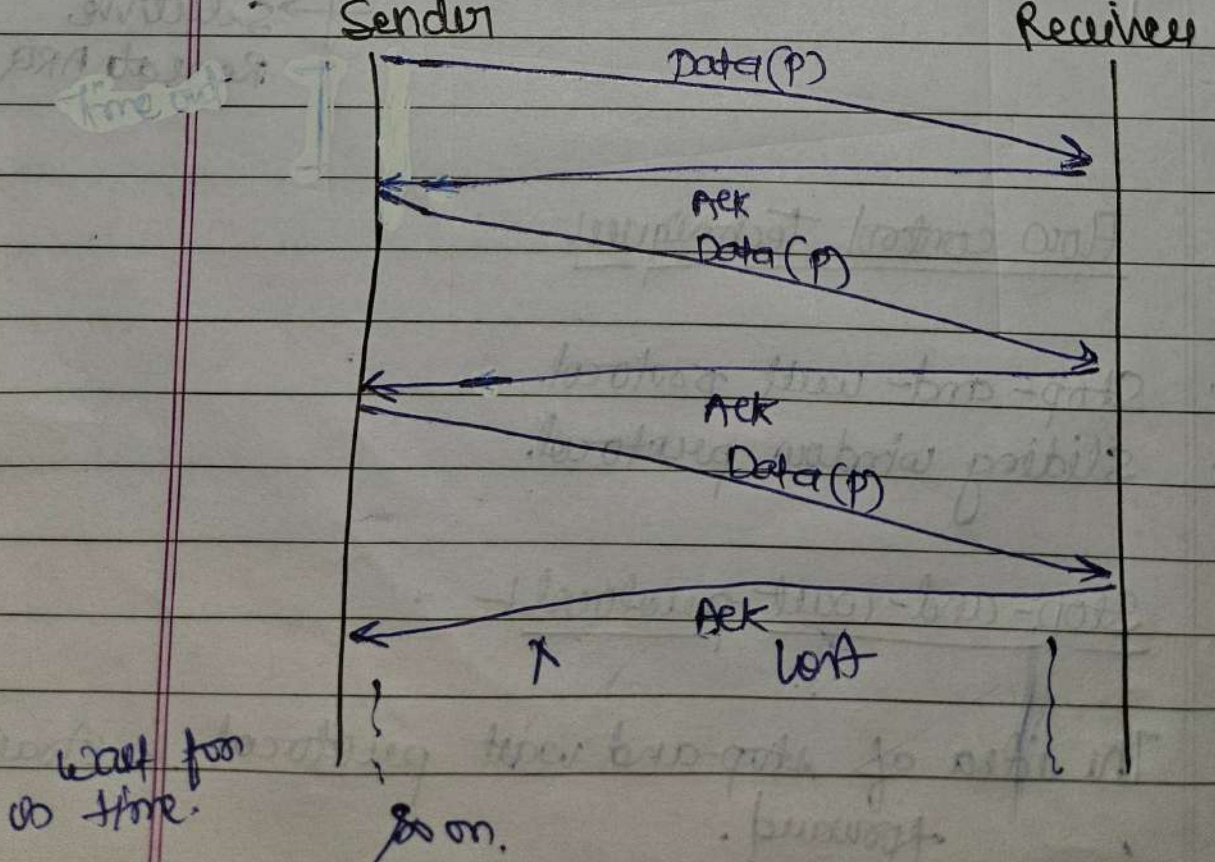
⊙ <u>Primitives of stop-and-wait Protocol</u>

1. <u>Sender side :-</u>

Rule 1. Send one data packet at a time.
Rule 2. Send the next packet only after receiving ACK for the previous.

2. <u>Receiver side :-</u>

Rule 1. Receive and consume data packet.
Rule 2. After consuming packet, Ack need to be sent (Flow control)



⊙ problems:-

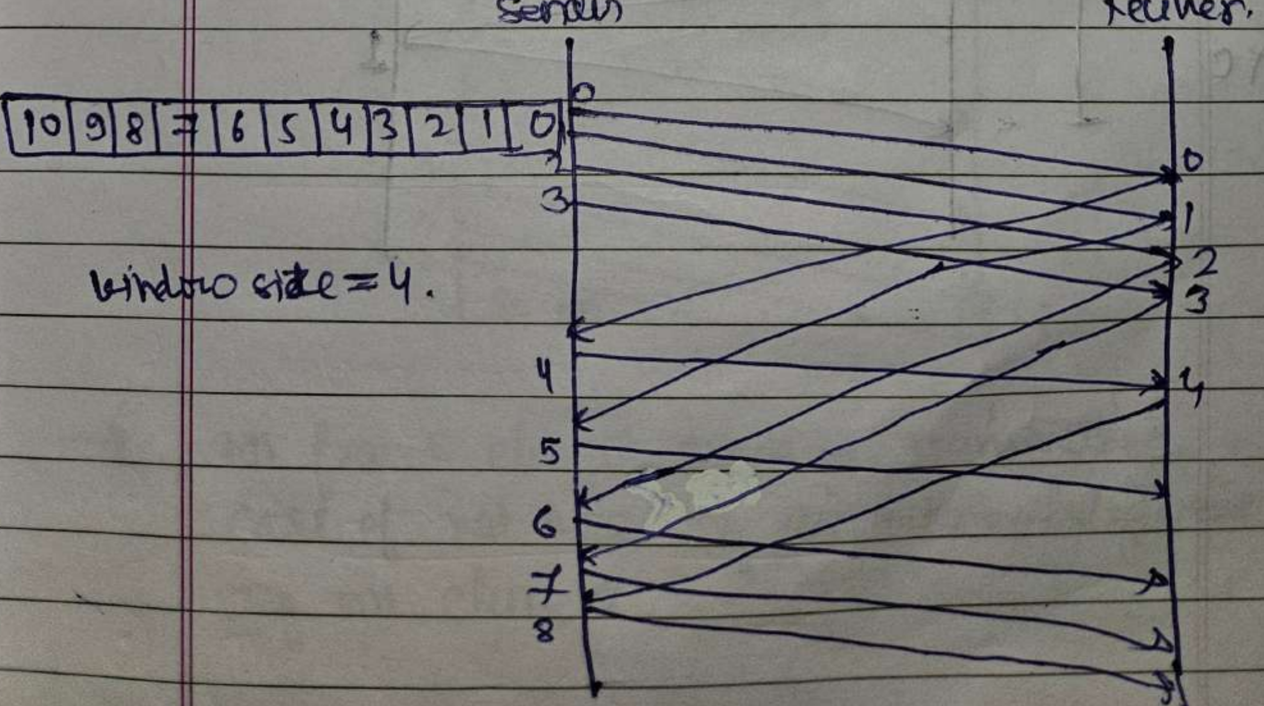
1. Problem due to lost data,
∴ Sender waits for ack
Receiver waits for data.

2. problem due to lost Acknowledgement,
∴ Sender waits for ack, an infinite amount of time.

3 Problem due to delayed Ack/data.

⊙ Sliding window protocol

⊙ Allows sender to send multiple frames before needing an Acknowledgement.
→ Maintain a window of frame that can be sent.

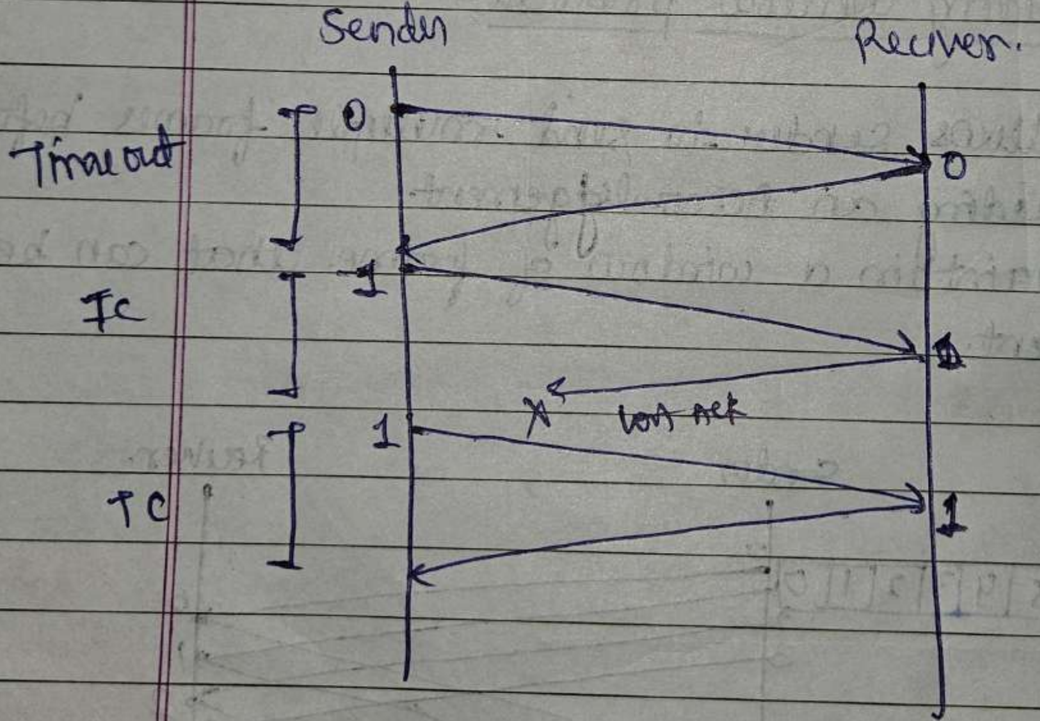


Stop-and-wait-ARQ:-

- After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.

→ If the acknowledgement does not arrive after a certain period of time the sender times out and retransmits the original frame.

Stop-and-wait-ARQ = stop and wait + Timeout + sequence number.



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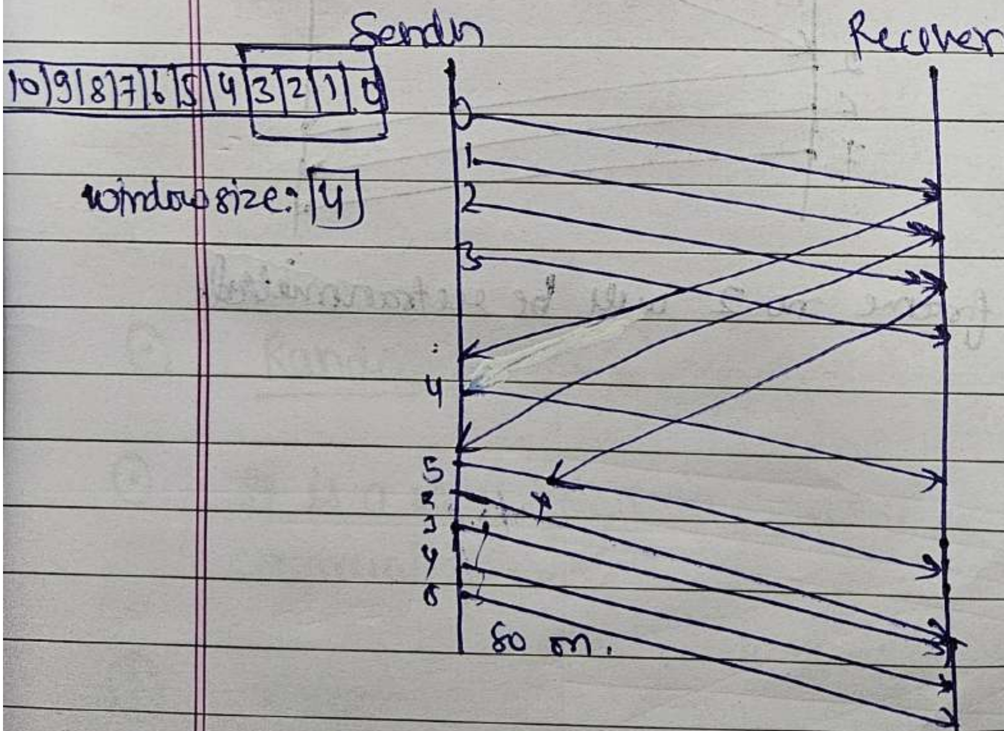
⊙ GO-Back-N-ARQ :-

⊙ GO-Back-N-ARQ
↳ N is the sender window size.

→ It uses protocol pipelining.

→ There are finite numbers of frames and the frames are numbered in a sequential manner.
→ Even for one acknowledgement, all frames in the current window are transmitted.

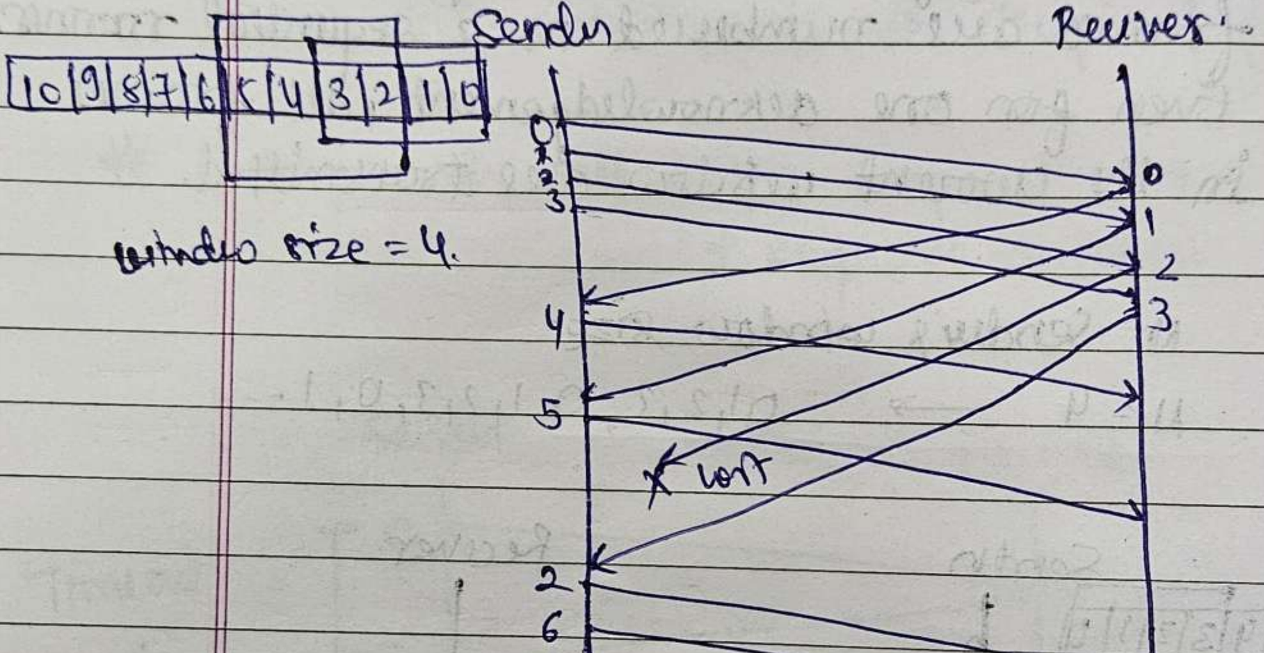
→ N- Sender's window size.
→ N = 4 ⟶ 0,1,2,3,0,1,2,3,0,1...



→ All frame element should be transmitted, in the case of not receiving the Acknowledgement from any one element.

⊙ <u>selective-repeat ARQ:</u>

⊙ In this only the lost frames are retransmitted.



window size = 4.

only frame no 2 will be retransmitted.

Medium Access Sub Layer

⊙ Medium Access sub layer:-

⊙ Medium Access sub layer is a part of data link layer in OSI model.

→ It is responsible for controlling how multiple devices share and access the physical transmission medium in an efficient and collision-free manner.

Medium Access protocols (MAC)

- Random Access protocol
  - Aloha
  - CSMA
  - CSMA/CD
  - CSMA/CA
- Control Access protocol
  - Polling
  - Token polling
- Channelization Access protocol
  - FDMA
  - TDMA

⊙ Random Access protocol:-

⊙ It is a way for many devices to share the same communication line.

(1) Aloha:-

⊙ A way for devices (like computers or phones) to send data without checking if others are also sending.

→ LAN based protocol.

→ Send data whenever you want.
→ If two devices send at the same time, the data gets lost (collision).
→ Wait a bit and try sending again.

ALOHA
→ pure Aloha
→ Slotted Aloha.

⊙ Pure Aloha :-

⊙ LAN Based.
→ only transmission time.
→ No propogation time.

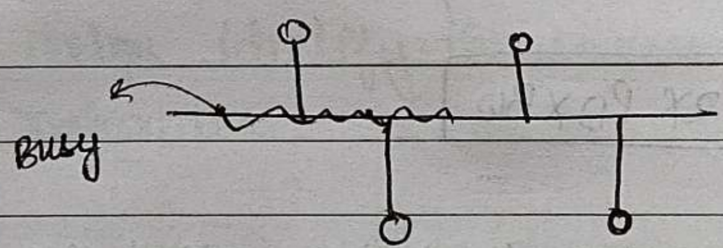
1. Vulnerable time.
2. Efficiency.

CSMA :-

- Carrier - sensor Multiple Access

  - 1 - persistant. (check again and again).
  - 0 - persistant. (wait for some random time).
  - p - persistant. (combination of both).

→ value of probability.



→ If collision occurs bandwidth - waste.
throughput - down.

→ It does not sense the entire medium.

③ CSMA / CD :-

⊙ Carrier-Sence Multiple Access / collision Detection

→ Transmission time > propogation Delay

⇒ Transmission time ≥ 2 × Propogation Delay.

$$\boxed{TT \geq 2 \times PD} \rightarrow \boxed{\frac{D}{V}}$$

⇒ $TT = \frac{L}{BW}$ → length of message. BW → Bandwidth.

⇒ $\frac{L}{BW} \geq 2 \times PD$

⇒ $\boxed{L \geq 2 \times PD \times BW}$

Que. Consider a CSMA/CD ~~that~~ transmitted data at a rate of 100 Mbps (10^8 bits per second) over 1 km cable with no repeated. Min frame size required is 1250 bytes. What is the signal speed (km/sec) in the cable?

⇒ Given :- $D = 1\,km$

$BW = 100$ Mbps. (10^8 bits)

$L = 1250$ bytes × 8 bits.

⇒ $V = \frac{2 \times 1 \times 10}{1250 \times 8}$

⇒ $\boxed{V = 2 \times 10^4}$

Network Layer

⊙ Network layer:-

⊙ The Network layer is responsible for logical addressing, routing and forwarding of data. It decides how data travels across network and ensure it reaches the correct destination.

⊙ IP addressing:-

⊙ An IP address is a unique number assigned to every device connected to a network.

→ IP → Internet protocol.

→ Helps identify the source and destination across networks.

⇒ 32 bits $= 2^{32}$ addresses.
↳ 4294967296.

⇒ 128 bits $= 2^{128} \cong \infty$.

⊙ Types → IPV4, IPV6

1. IPV4:-

→ uses 32 bits (4 octets).

→ written as four numbers separated by dots.

3. Each numbers range from 0 to 255.
4. Example! 192.168.1.1
5. Total possible addresses = 2^{32} = 4294967296

(2.) IPV6 :-

1. uses 128 bits (much larger address space)
2. written as 8 groups of hexadecimal numbers seperated by colon.
3. Ex! 2001:0db8:8593:0000:0000:892e:0370:7334
4. Total possible addresses: 2^{128} = (almost unlimited)

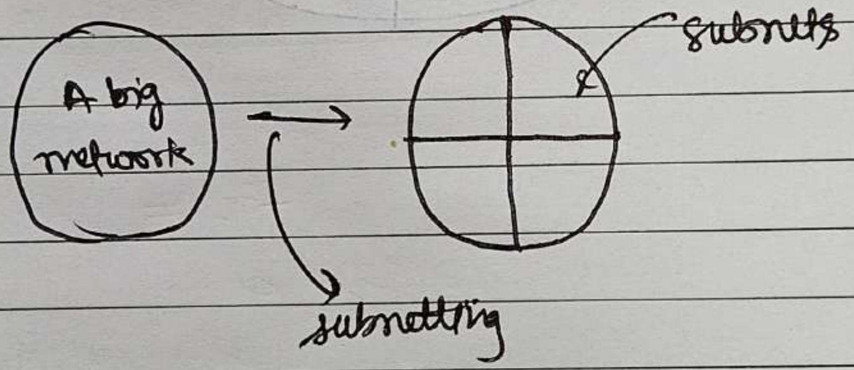
NOTE: universal and unique.

• Submetting :-

⊙ Submetting is a process of dividing a large network into smaller logical networks (subnets).

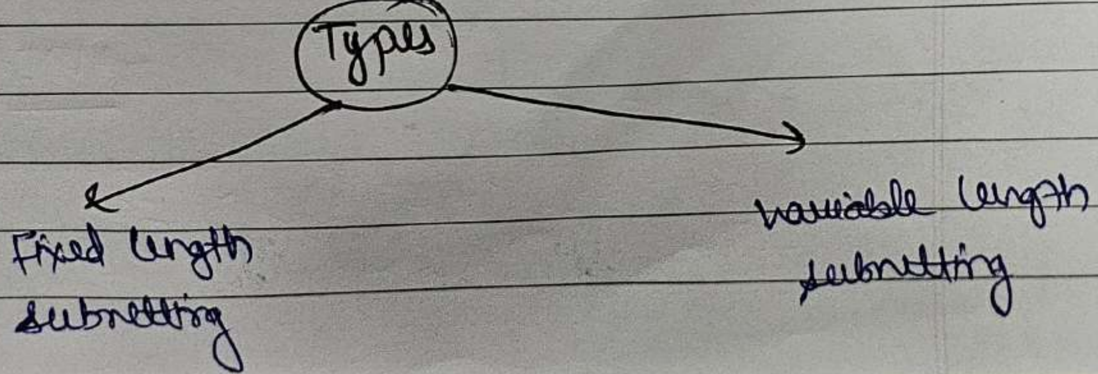
Advantages:-

1. To use IP addresses efficiently.
2. To improve network performance.
3. To enhance security.
4. To reduce broadcast traffic.



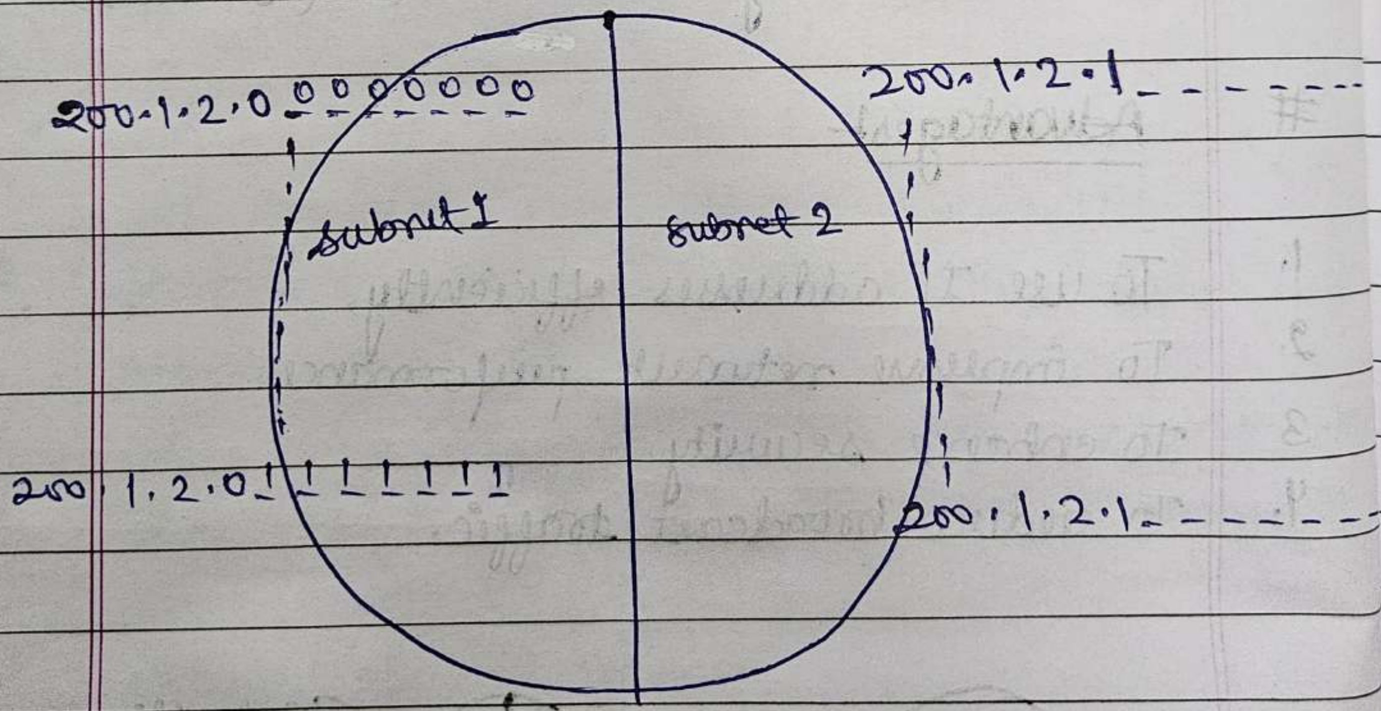
Disadvantages:-

1. Identification of a station is difficult.
2. Not possible to directed broadcast from outside network.



Fixed-size/length Subnetting

Ques. Consider the network having IP Address 200.1.2.0 Divide this network into two subnets.

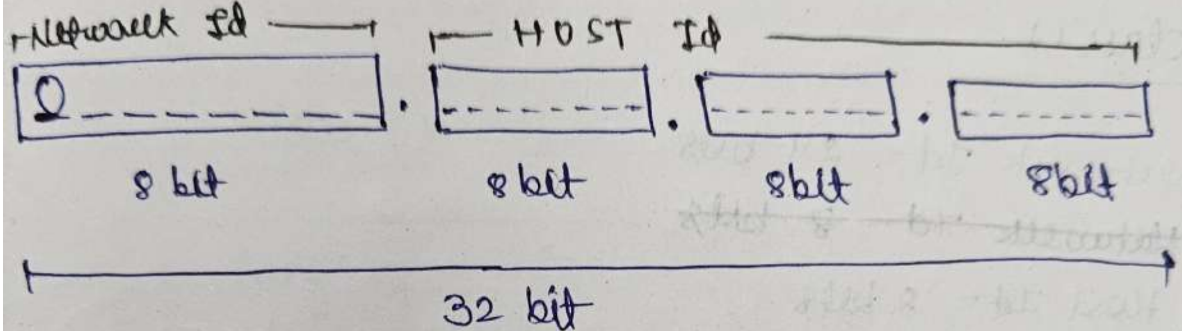


Classfull Addressing in Detail :-

- In this method, IP addresses are divided into fixed size based on the first few bits of the address.

1. class A :-

→ Net ID = 8 bits
→ HOST ID = 24 bits
→ First bit is reserved to 0 in binary notation.



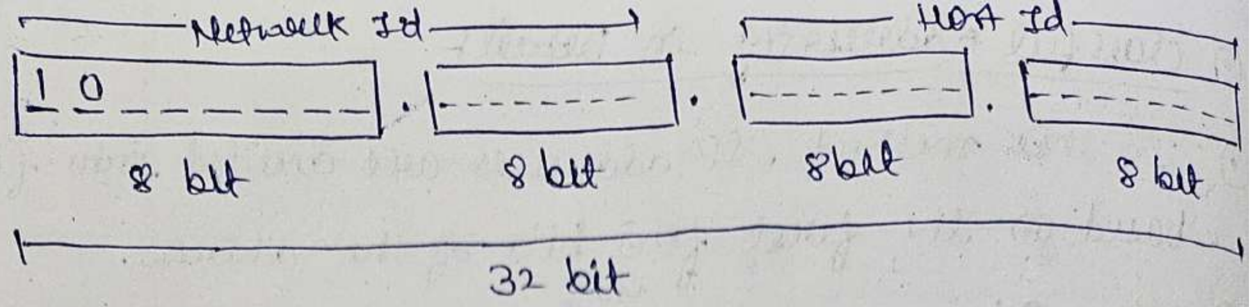
→ No. of Networks = $2^7 \Rightarrow 128$.
→ 00000000 and 01111111 are not use by any org. so, total possible that are used are, 126.

→ No. of Host = 2^{24}
→ 64.0.0.0 and 64.255.255.255 are not given to any host. so, the Host are = $2^{24}-2$

→ range = 0 – 127
→ Default mask = 255.0.0.0

2. class B :-

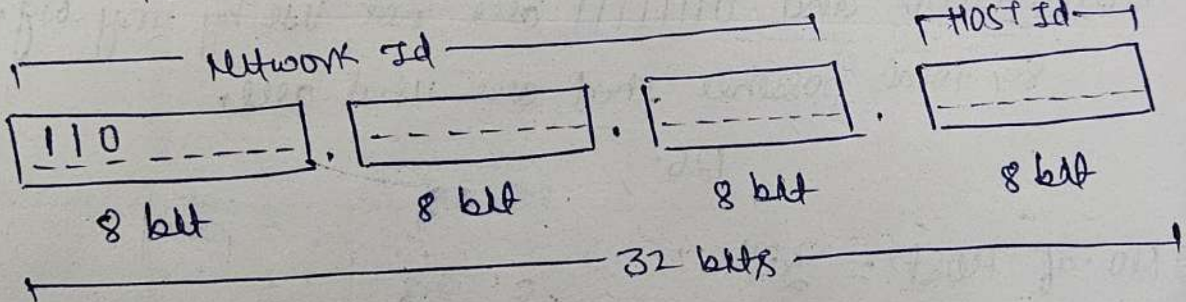
→ Network Id = 16 bit
→ HOST Id = 16 bit
→ Start bits 10.



→ Range = 128 − 191
→ No. of Network = $2^{16} - 2$
→ No. of Host = $2^{16} - 2$

③ Class C:

→ Network Id = 24 bits
→ Host Id = 8 bits
→ Range = 192 − 223
→ First 3 bits are 110



→ No. of Networks = $2^{24} - 2$
→ No. of Host = $2^{8} - 2$

④ Class D:

→ Network Id = NA
→ Host Id = NA
→ Range = 224 − 239
→ Starts with = 1110

→ Multicasting Group Email / Broad cast (Reserved)

1 1 1 0 - - - -	- - - - - - - -	- - - - - - - -	- - - - - - - -
8 bits	8 bits	8 bits	8 bits

32 bits

→ class D is not divided into Network Id and HOST Id.

5. **Class E:-**

→ Range → 240 – 255
→ starts with → 1111

1 1 1 1 - - - -	- - - - - - - -	- - - - - - - -	- - - - - - - -
8 bits	8 bits	8 bits	8 bits

32 bits

→ Research, Future use (Reserved).

⊙ **Disadvantages:-**

1. IP address wastage.
2. No flexibility.
3. routing tables become large.
4. IPV4 exhaustion becomes faster.

→ Classfull addressing is an older method where IP addresses were divided into fixed classes like A, B, C, D and E.

→ This cause a lot of IP wastage because network couldn't choose the exact number of address they needed.

→ So, classless addressing (CIDR) was introduced to give more flexibility, allowing networks to use only the number of IP's they actually required and reducing wastage.

Conclusion of Classfull addressing:-

Class	Prefixes	First byte
A	n = 8 bits	0 to 127
B	n = 16 bits	128 to 191
C	n = 24 bits	192 to 223
D	Not Applicable	224 to 240
E	Not Applicable	240 to 255

Variable-Length Subnetting

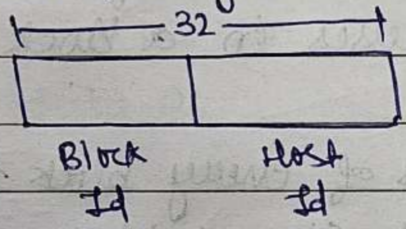
1. Classless Addressing:-

- classless addressing, also called CIDR (Classless Inter-Domain Routing), is the modern IP addressing method that does not use classes (A, B, C).

Instead, it uses prefix length (like /24, /20, /28) to define bits for Network and Host.

→ comen in 1993.

→ No classes / only blocks



→ **Notation:-**

$x.y.z.w/n$ → no. of bits represent block/network

Ex.

1. 200.10.20.40/28 = 200.10.20.0010|1000 (28 | H Id)

means = Net Id = 28
Host Id = 16

2. A.B.C.D/20

= Net Id = 20
= Host Id = 12
= Number of hosts = $2^{12} = 4096$

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Advantages:

1. No Id wastage.
2. Supports subnetting & supernetting
3. Smaller routing tables.
4. works for ISPs & enterprises.

Rules:

1. Addresses should be contiguous.
2. No. of addresses in a block must be in power of 2.
3. First address of every block must be evenly divisible with size of block.

Example 1. 200.10.20.40/28

Answer 1. 200.10.20.00101000

convert host bit to zero

= 200.10.20.00100000

= 200.10.20.32/28 (Network Id)

= Host bits = 4. ⇒ 2^4 ⇒ 16.

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(•) Subnetting in CIDR :-

195.10.20.10000000 → Fix
195.10.20.10000000
00001
00010
00100
S2
S1
195.10.20.10111111

195.10.20.10100000
195.10.20.10111111
160 to 191
195.10.20.128/26

195.10.20.128 to
195.10.20.159

$$195.10.20.128/26 = 195.10.20.\underbrace{10}_{N\,Id}\underbrace{000000}_{H\,Id}$$

So, = Net Id ⇒ 26
= Host Id ⇒ 6

32	32
− 2	− 2
30	30 — usable

S1	S2
195.10.20.128/27	195.10.20.160/27
to	to
195.10.20.159/27	195.10.20.191/27

(•) Numerical on CIDR :-

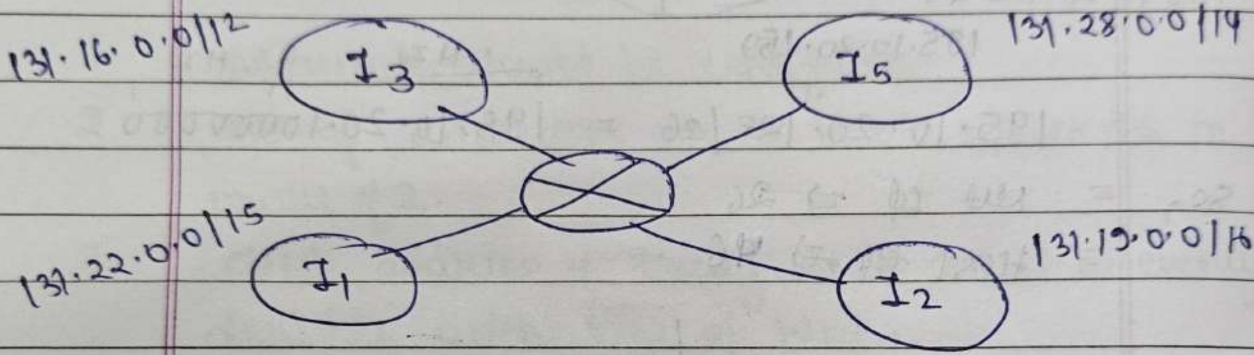
Ques! classless inter domain routing receive a packet with address 131.23.151.76 . The router's routing

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

table has following entries:

prefix	output interface
131.16.0.0/12	3
131.28.0.0/14	5
131.19.0.0/16	2
131.22.0.0/15	1

packet will be forwarded to which interface?



Given :- 131.23.151.76

(i) 131.16.0.0/12 ⟶ Net Id = 12

Host Id = 20

⇒ 131.23.151.76

⇒ 131.00010111.151.76

AND ⇒ 131.11110000.00000000.00000000

131.00010000.0.0

⇒ $\boxed{131.16.0.0}$ ✓

(ii) 131.20.0.0/14

Net Id = 14

Host Id = 18

131 · 23 · 151 · 76
131 · 00010111 · 151 · 76
AND 11111111 · 11111100 · 00000000 · 00000000
131 · 00010100 · 0 · 0

$\boxed{131 \cdot 20 \cdot 0 \cdot 0}$ X

(iii) 131 · 19 · 0 · 0 | 16

Net Id = 16

Host Id = 16

⇒ 131 · 23 · 151 · 76

AND ⇒ 131 · 00010111 · 151 · 76

11111111 · 00010011 · 11111111 · 11111111

131 · 00010011 · 151 · 176

⇒ $\boxed{131 \cdot 19 \cdot 0 \cdot 0}$ X

(iv) 131 · 23 · 151 · 76

AND 131 · 00010111 · 151 · 76

11111111 · 00010110 · 11111111 · 11111111

131 · 22 · 0 · 0

⇒ $\boxed{131 \cdot 22 \cdot 0 \cdot 0}$ ✓

⇒ choose the answer where the no. of one's are more in prefix.

So, Interface 1 is the Answer.

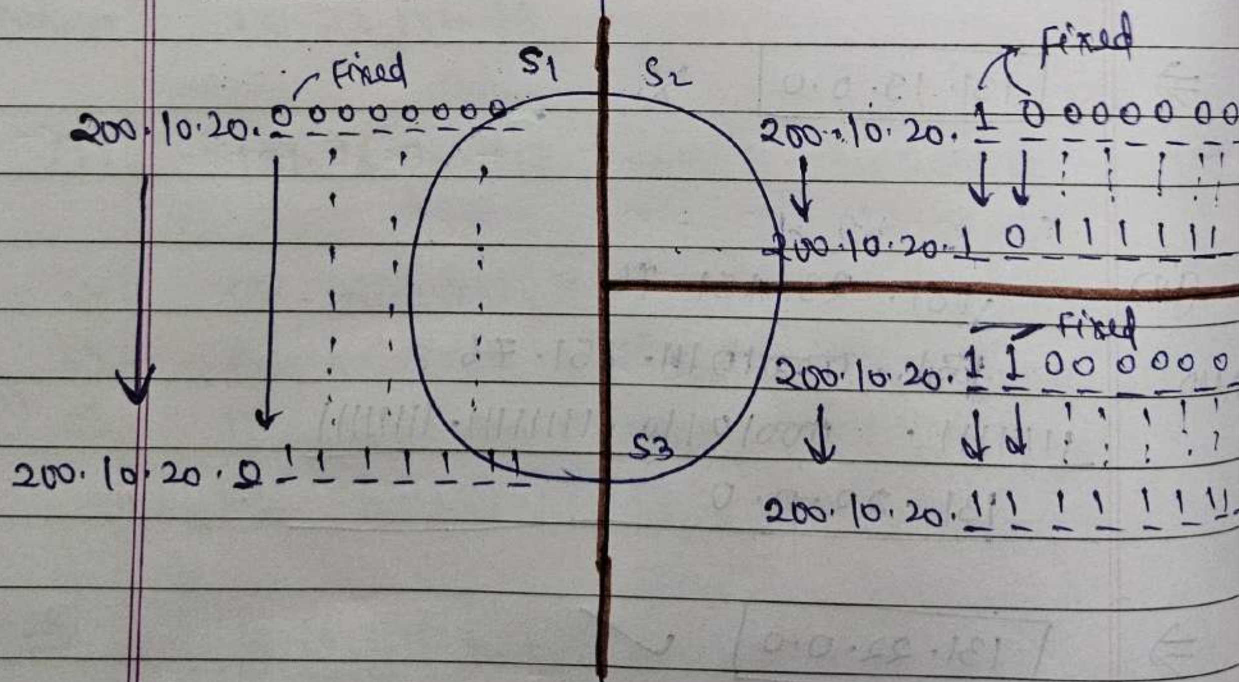
(2) (i) **Variable length subnet masking :-**

(i) VLSM is a subnetting technique where different subnets can have different subnet mask with in the same network.

(i) **Advantages :-**

1. Efficient IP utilization.
2. Reduces wastage.
3. Supports hierarchical addressing.
4. Better route summarization.
5. works perfectly with CIDR (classless).

(i) **Subnetting :- (classfull)**



=) Subnet I.

Range :- 200.10.20.0 to 200.10.20.127

usable = 128 - 2 = $\boxed{126}$

=) Subnet 2:

Range: 200.10.20.128
to
200.10.20.191

usable = 62 - 2
= $\boxed{60}$

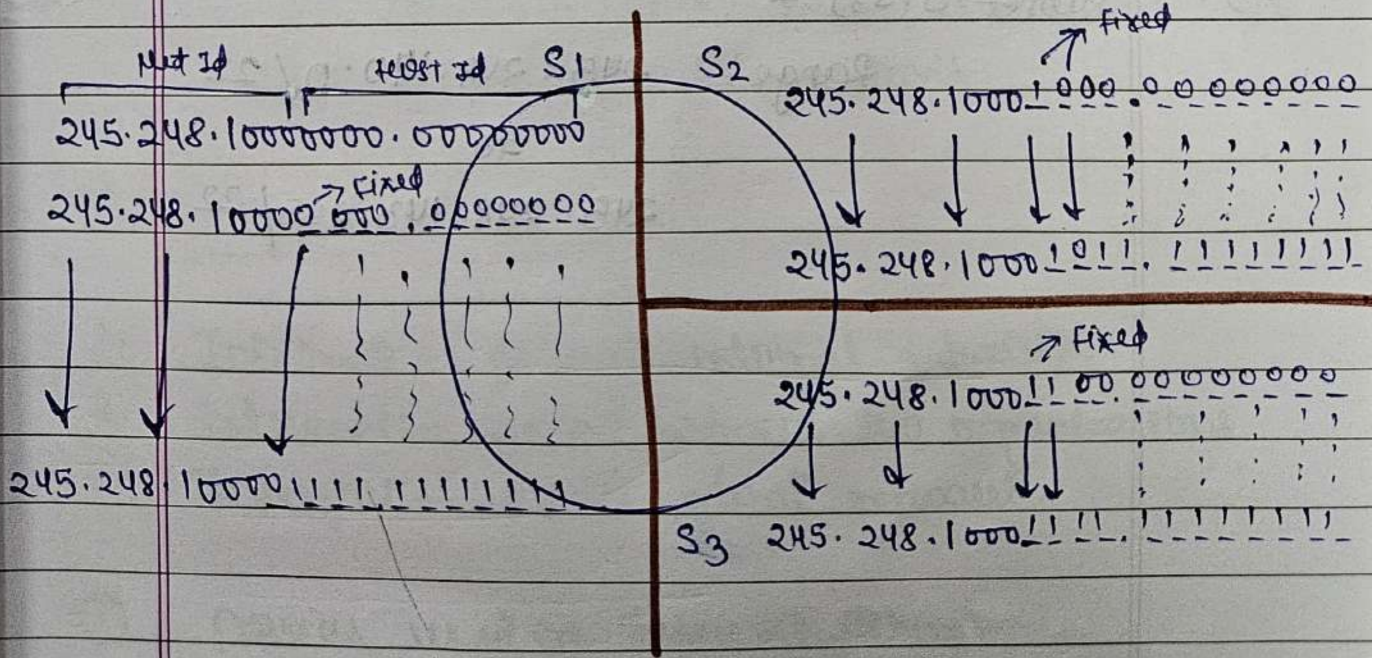
=) Subnet 3:

Range: 200.10.20.192
to
200.10.20.255

usable = 62 - 2
= $\boxed{60}$

=) Total usable = 256 - 6
= $\boxed{250}$

⊙ Subnetting in CIDR (VLSM): (classless)



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=) 245.248.128.0/20

=) Net Id = 20
=) Host Id = 12

=) subnet 1 (S_1):
Range = 245.248.128.0/21
to
245.248.135.255/21

=) subnet 2 (S_2):-
Range = 245.248.136.0/22
to
245.248.139.225/22

=) Subnet 3 (S_3) :-
Range :- 245.248.140.0/22
to
245.248.143.255/22

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Internetworking :-

- Internetworking is the process or technique of connecting multiple network so they can communicate and share resources.

→ It is also known as a network of networks

- Ex:-

LAN + WAN + MAN + PAN. → form a combined connected network.

The largest internetwork in the world = The Internet.

Why Internetworking is needed :-

1. Share data between networks.
2. share resources
3. Allows communication across cities / countries
4. Expand networks without limit.

Types:-

1. Intranet :- private internal network.
2. Extranet :- shared between two organizations
3. Internet :- public global network.

Devices used in Internetworking :-

1. Repeater

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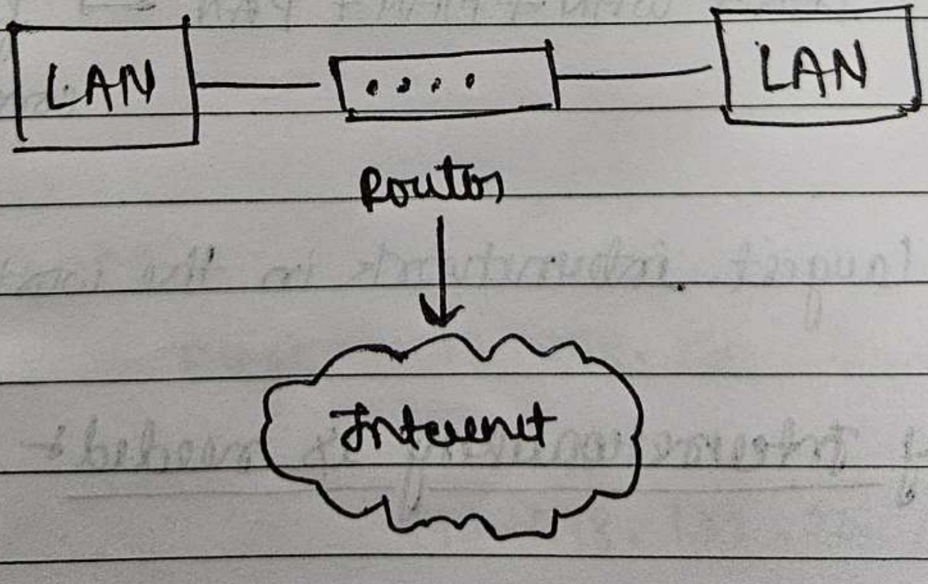
2 Hub.
3. Switch.
4. Bridge.
5 Router.
6. Gateway.
7 Modem.



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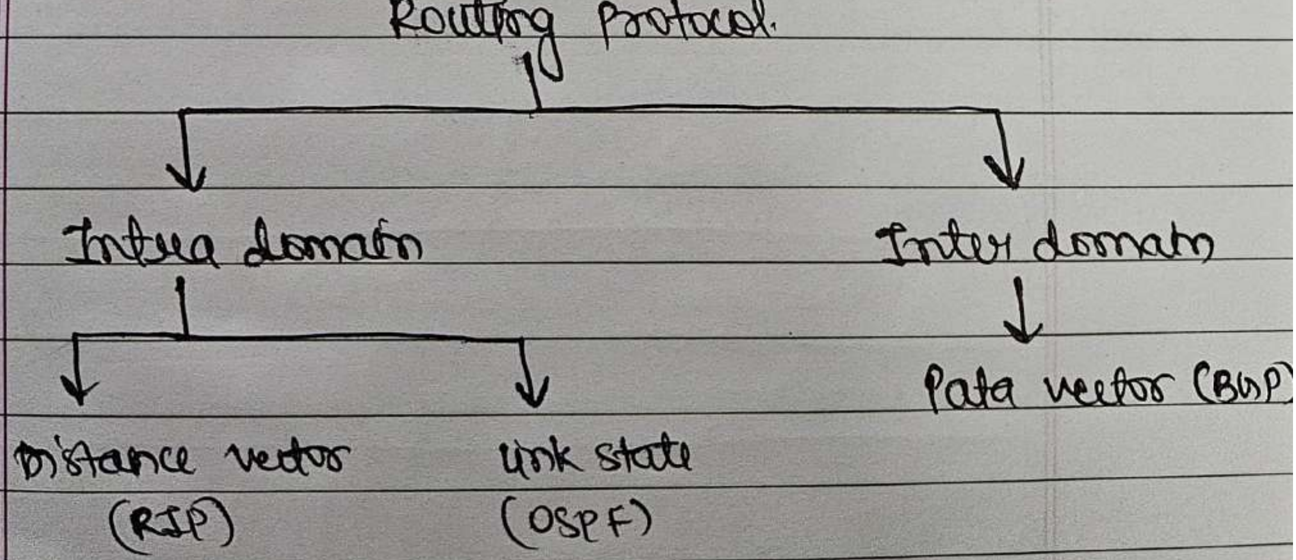
Interdomain and Intradomain Routing (RIP, OSPF and BGP) :—

Routing protocol:

A routing protocol is a set of rules used by routers to automatically finds the best path for sending data from source to destination.

Why Routing protocol are needed:

1. Discover other routers.
2. Learn network topology.
3. Select best path.
4. update Routing table automatically.



Transport Layer

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- ## Transport Layer

- The Transport Layer (Layer 4 of OSI model) is responsible for process-to-process communication

⇒ end-to-end delivery of entire message, addressing, reliable delivery, flow and error control

protocol

UDP — user datagram protocol

TCP — transport control protocol

- ## Design Issues

1. Induced Traffic.
2. Power and bandwidth constraints.
3. Throughput unfairness.
4. Interpretation of congestion.
5. Dynamic topology.

- ## TCP:-

- Transport control protocol.
- TCP is most widely used for data transmission in communication network such as Internet. TCP provides process-to process communication using PORT numbers.

←20-60→

Header	Data

⊙ TCP Headers

⊙ A TCP header is typically 20 bytes (minimal) and contains various fields used for reliable communication.

⊙ Features
- → Byte streaming
- → connection oriented → 3 way, → 4 way
- → full Duplex
- → Piggybacking → Go-Back-N, → selective - Repeat
- → Error control
- → Flow control
- → congestion control

→ Range = (20-60) B
↓
(160 bits — 480 bits)

Header Format

<table>
<tr><td colspan="5">source port SP (16)</td><td colspan="5">DP (16) destination port</td></tr>
<tr><td colspan="10">Sequence Number (32)</td></tr>
<tr><td colspan="10">Acknowledgement Number (32)</td></tr>
<tr><td>HL (4)</td><td>R (6)</td><td>URG</td><td>ACK</td><td>PSH</td><td>RST</td><td>SYN</td><td>FIN</td><td colspan="2">window size (16)</td></tr>
<tr><td colspan="5">checksum (16)</td><td colspan="5">urgent Pointer (16)</td></tr>
<tr><td colspan="10">Data</td></tr>
</table>

Application Layer

↓ Data / message $\boxed{M}$

Transport Layer

↓ add TCP header to message comes from Application layer $\boxed{H \mid M}$

Session Layer

TCP Segment

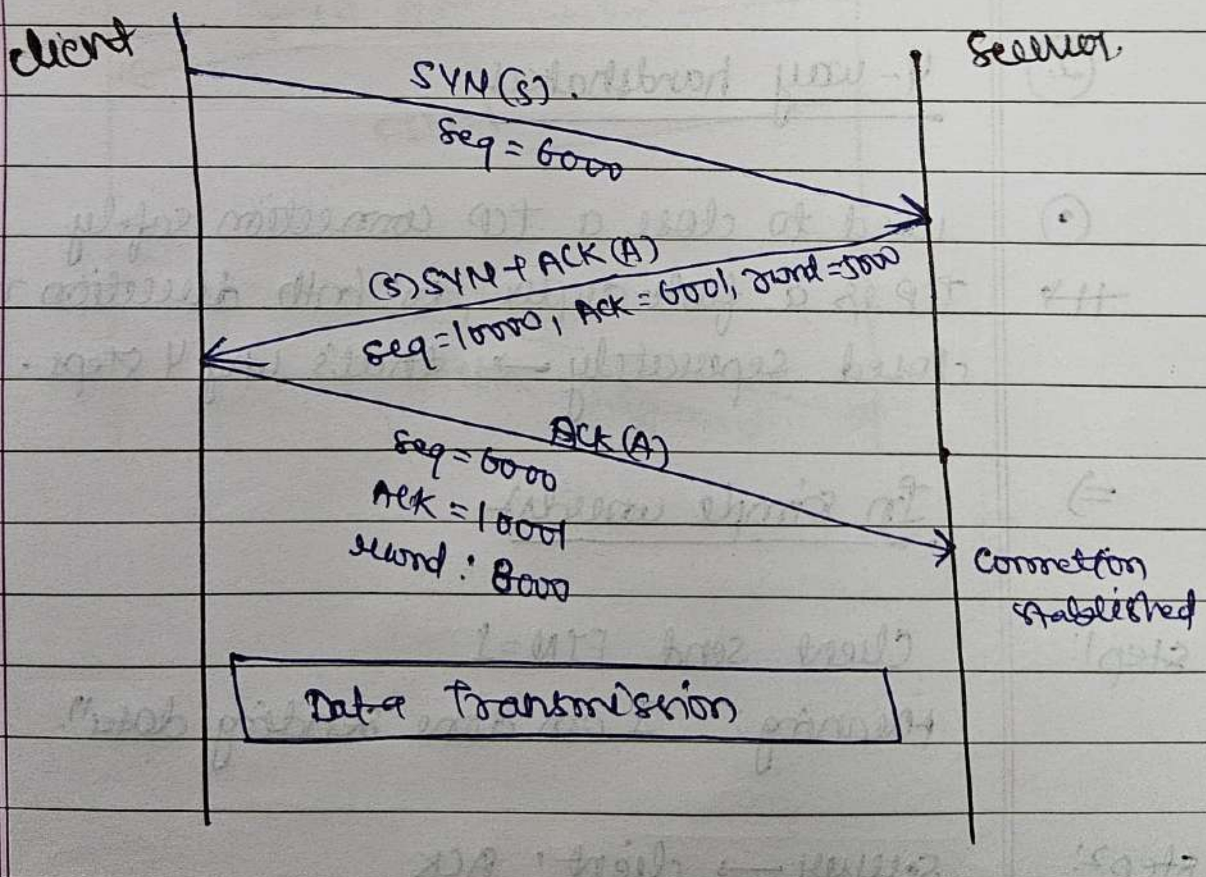
Connection Management

- TCP uses special steps to establish and terminate a reliable connection.

Types → 3-way Handshake, 4-way handshake.

(1) 3-way Handshake

- Used to create a TCP connection between client and server.



⇒ In simple words:

(i) step 1: client ⟶ server (SYN)

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client sends a packet with SYN = 1
Meaning: "I want to start a connection"

(ii) step 2:- Server → client: SYN + ACK
Server replies with SYN = 1 and Ack = 1
Meaning:
"I accept your request".
"Let's synchronize sequence numbers".

(iii) step 3:- client → server: Ack
client sends final Ack
Meaning: "connection established".

(2) 4-way handshakes

(•) used to close a TCP connection safely.
TCP is a full duplex, so both direction must be closed seperately → that's why 4 steps.

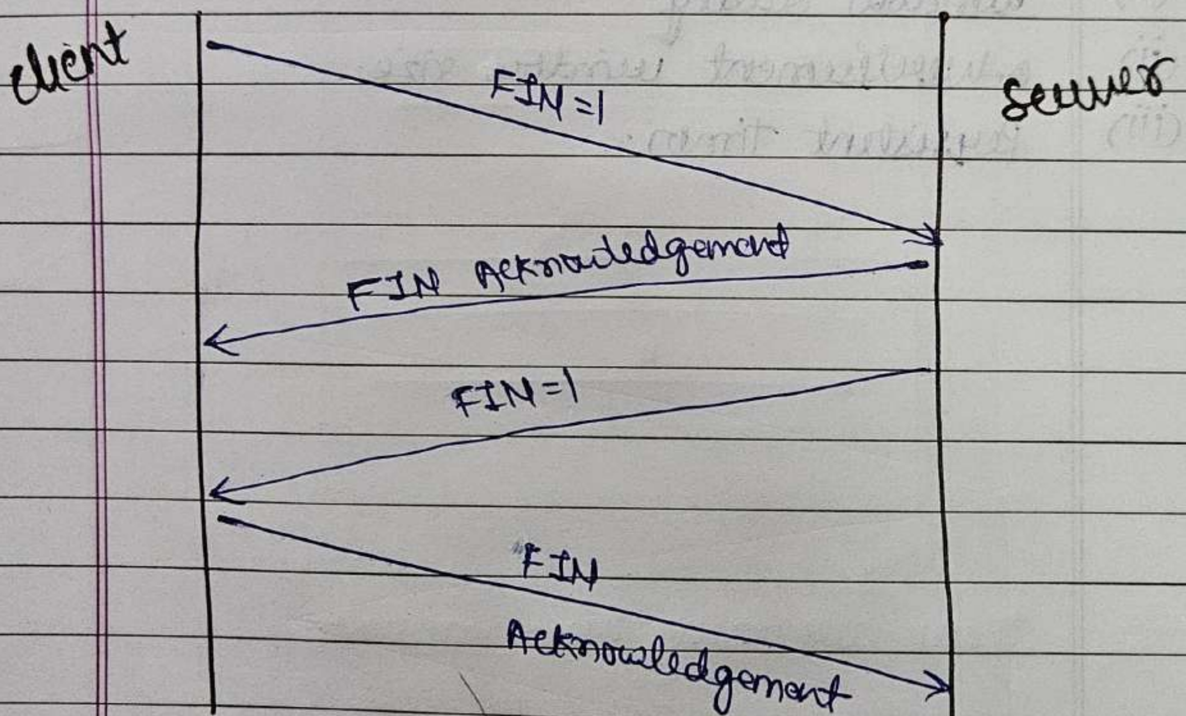
=) In simple words:

step 1: Client send FIN = 1
Meaning: "I am done sending data".

step 2: server → client: ACK
Server sends ACK
Meaning: "I received your FIN".

Step3: Server → client : FIN
when server is ready to close
server sends FIN=1
Meaning: "I am also done sending data".

Step4: client → server : Ack
client sends final Ack.
Meaning : "connection closed".



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⊙ Flow control:-

⊙ Flow control explains how transfer speed of data is adjusted in between two computer during data transmission.

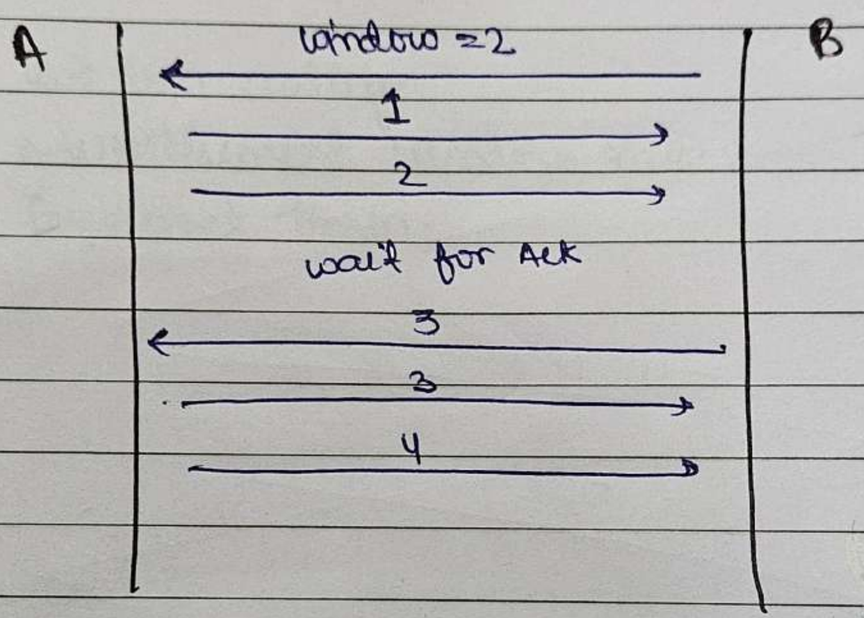
→ In TCP, flow control is maintained by following key tools:

(i) window scaling.
(ii) Advertisement window size.
(iii) Persistent Timer.

TCP window management

TCP uses a window mechanism to control how much data can be sent without waiting for an acknowledgement.

This ensures reliability, flow control, and congestion control.



→ There are three main windows

1. Send window.
2. Receive window. (rwnd)
3. Congestion window. (cwnd).

(i) Send window

1. How much data the sender is allowed to send depends on

send window = min(rwnd, cwnd)

Sender cannot send more that this limit.

(2) Receive window (rwnd):-

→ Free buffer space at the receiver
→ used to flow control.
→ If rwnd = 0 ⟶ sender must stop sending.

(3) Congestion window (cwnd):-

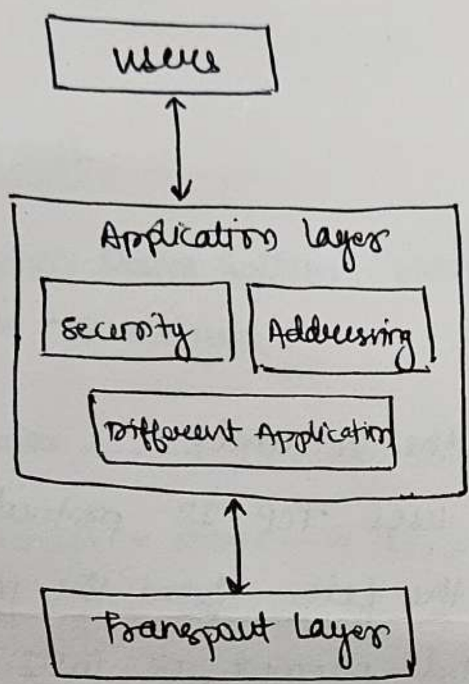
→ limit based on network congestion, not receiver.
→ used for congestion control.
→ Increase when network is free, decrease when congestion occurs.

Application Layer

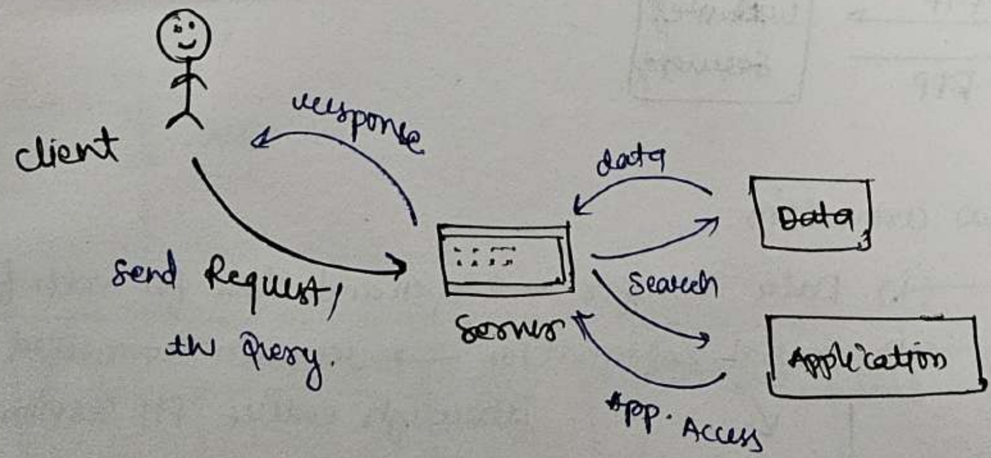
- The Application Layer is the topmost layer of the OSI model and provides services directly to end-users.

→ It enables network applications like email, web browsing, file transfer video streaming etc.

→ allows people to use the internet.



→ Application layer programs are based on the client-server model.



→ convert data into human-readable form.

→ www.

→ Multimedia.

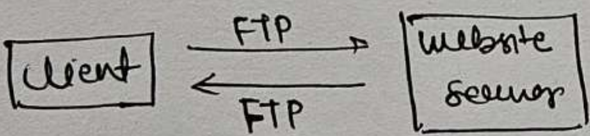
⊙ Important Protocols:-

1. FTP — File transfer.
2. DNS
3. HTTP | HTTPS
4. SMTP
5. TELNET
6. POP3 / IMAP.
7. DHCP

(1.) FTP :-

⊙ File transfer protocol.

→ It is used to exchange files on the Internet.
→ To enable data transfer FTP uses TCP | IP protocol.
→ used to uplode and download the files from the internet.
→ can be invoked from command prompt or GUI.
→ allow update, delete, rename, move and copy files at a server



FTP established two connection

(i) Data Transfer. → open and close for each file

→ used for sending the actual file.

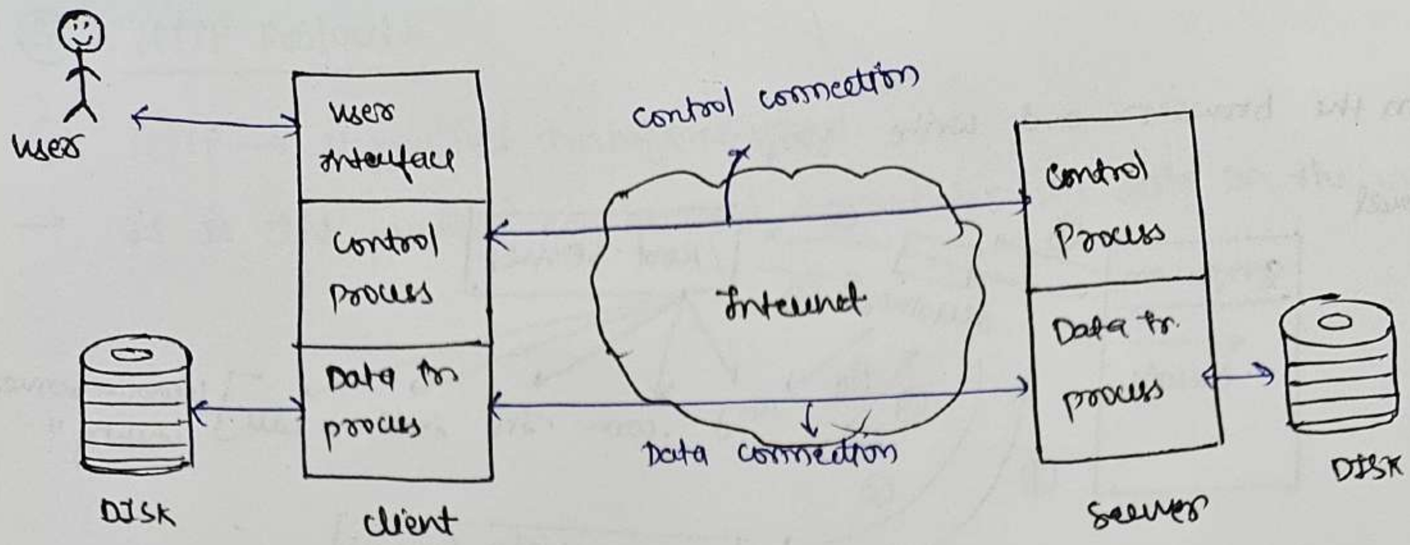
→ only for file transfer

(ii) Control Information → remain connected through entire FTP session

→ used for talking between client and server

→ sends command like:-

1. login.
2. show.
3. upload.
4. download.



DNS protocol

- DNS (Domain Name system) protocol is the rule that helps ^to convert^ a domain name ⟶ IP address

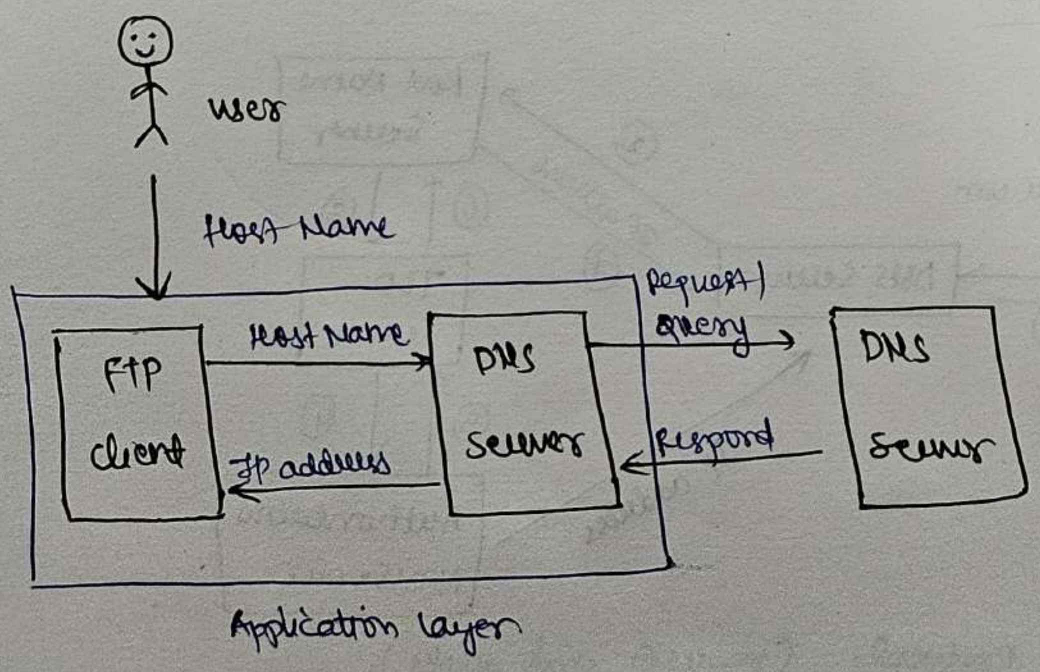
Ex: google.com ⟶ 142.250.182.206.

DN ⟶ DNS ⟶ IP Address.

DNS protocol = Name ⟶ Number.

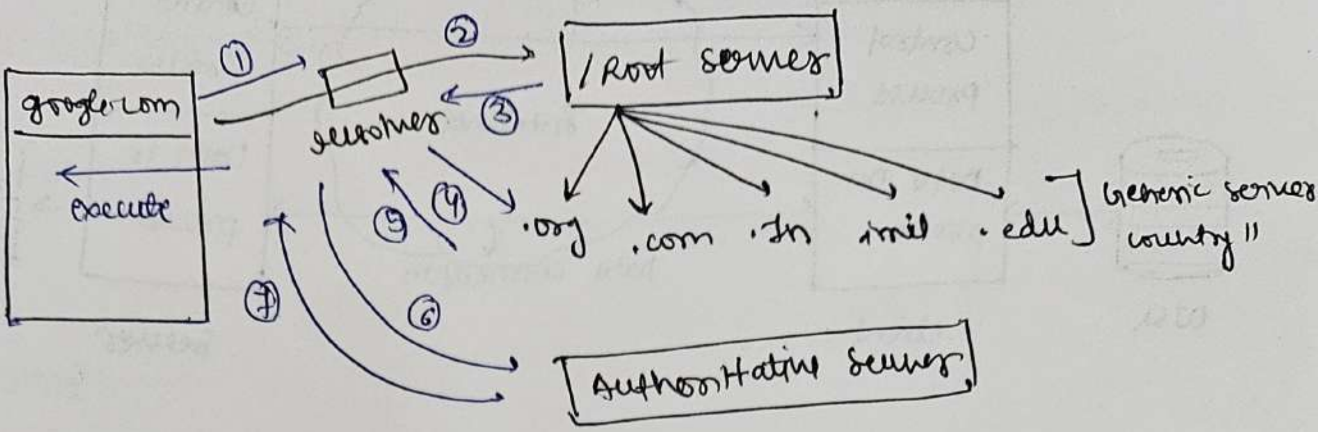
→ Because human remember website name, but computer only understand IP addresses.

DNS working:

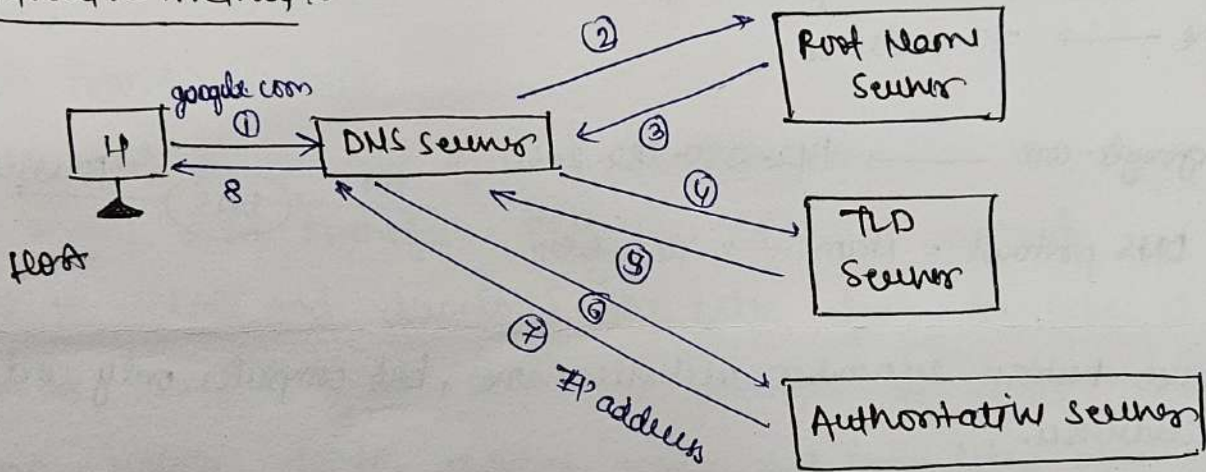


→ on the browser and write google.com.

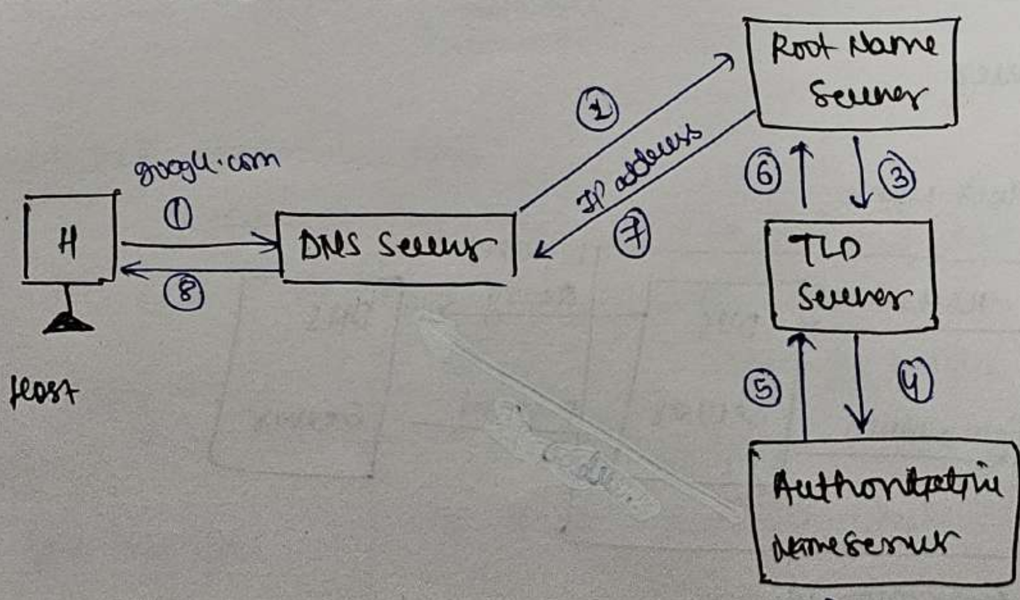
∴ General



① Iterative method :-



② Recursive Type :-



⇒ DNS use UDP protocol (request and reply)

③ <u>HTTP Protocol :-</u>

⊙ HTTP ⟶ Hyper text transfer Protocol.

→ It is the protocol can be used to transfer the data on the www.

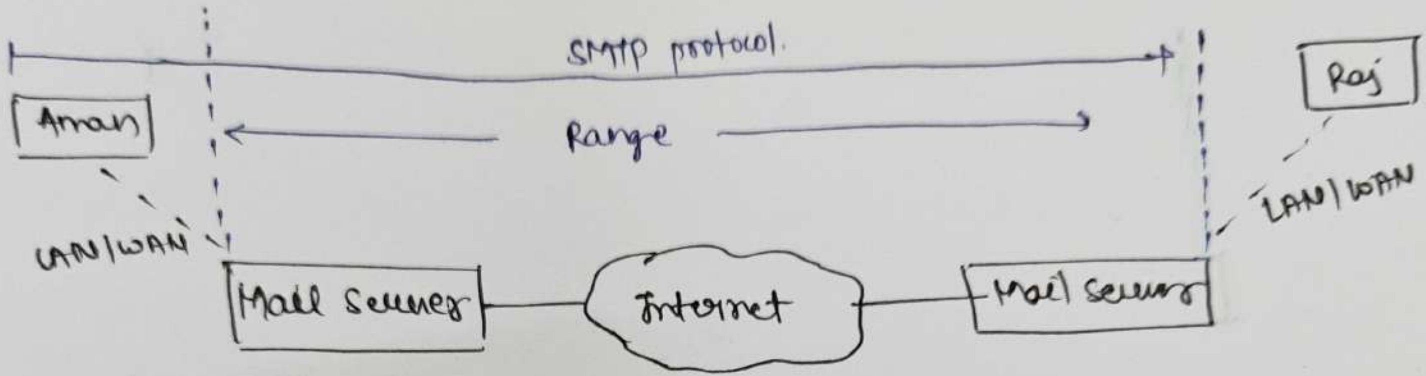
(4.) SMTP Protocol

- SMTP → Simple Mail Transfer Protocol

→ SMTP is a protocol used to send emails from one device to another device over the internet.

→ Used to send and Forwarding not receiving.

→ It is the protocol that defines MTA client and server in the internet



- Commands :-

1. HELO → used by client to identify itself.
2. MAIL FROM → Identify sender.
3. RCPT to → Identify intended recipient.
4. DATA → send actual message.
5. QUIT → QUIT → Terminate the message.
6. RSET → Reset the connection.
7. VRFY → verify the address of recipient.
8. HELP → HELP.